

Tab 1

Brandon's Notes

TODO:

- Understand how MST3R SLAM API folder works and how to run it
- How images are sent and processed

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10. Commands to Run for Each Step

Use these in order. Commands assume you are inside MAST3R-SLAM-API.

Step 0: Environment + dependencies

Install Conda first (Windows PowerShell):

```
```powershell
winget install -e --id Anaconda.Miniconda3
```
```

Close and reopen your terminal, then initialize Conda for PowerShell:

```
```powershell
conda init powershell
```
```

Note: Check if VSCode Powershell allows script execution

Run these commands to accept terms of service:

```
conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/main
conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/r
conda tos accept --override-channels --channel https://repo.anaconda.com/pkg/msys2
```

Close and reopen your terminal again, then verify:

```
```powershell
conda --version
```
```

If `conda` is still not recognized, run this once and reopen terminal:

```
```powershell
```

## Brandon's Notes

```
& "$env:USERPROFILE\miniconda3\Scripts\conda.exe" init powershell
...

also do that on VSCode terminal

Then create the environment and install dependencies:

```bash
conda create -n mast3r-slam python=3.11 -y
conda activate mast3r-slam

# Install PyTorch matching your CUDA version (example uses CUDA 12.1)
conda install pytorch==2.5.1 torchvision==0.20.1 torchaudio==2.5.1
pytorch-cuda=12.1 -c pytorch -c nvidia

pip install -e thirdparty/mast3r
pip install -e thirdparty/in3d
pip install --no-build-isolation -e .
pip install evo lpips scikit-image matplotlib
...

```

- Problem: The following command causes this error

```
python -m pip install --no-build-isolation -e
.\MASt3R-SLAM-API\thirdparty\mast3r
>>
Obtaining
file:///C:/Users/maste/OneDrive/Documents/Engineering/CSCE/Csce_483/TeamBa
llers/MASt3R-SLAM-API/thirdparty/mast3r
  Checking if build backend supports build_editable ... done
  Preparing editable metadata (pyproject.toml) ... done
Collecting scikit-learn (from MAST3R==0.0.1)
  Using cached scikit_learn-1.8.0-cp313-cp313-win_amd64.whl.metadata (11
kB)
Collecting roma (from MAST3R==0.0.1)
  Using cached roma-1.5.6-py3-none-any.whl.metadata (5.5 kB)
Collecting gradio (from MAST3R==0.0.1)
  Using cached gradio-6.10.0-py3-none-any.whl.metadata (17 kB)
Collecting matplotlib (from MAST3R==0.0.1)
  Using cached matplotlib-3.10.8-cp313-cp313-win_amd64.whl.metadata (52
kB)
Requirement already satisfied: tqdm in
C:\Users\maste\miniconda3\Lib\site-packages (from MAST3R==0.0.1) (4.67.3)
Collecting opencv-python (from MAST3R==0.0.1)
```

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```
Using cached opencv_python-4.13.0.92-cp37-abi3-win_amd64.whl.metadata
(20 kB)
Collecting scipy (from MAST3R==0.0.1)
  Using cached scipy-1.17.1-cp313-cp313-win_amd64.whl.metadata (60 kB)
Collecting einops (from MAST3R==0.0.1)
  Using cached einops-0.8.2-py3-none-any.whl.metadata (13 kB)
Collecting trimesh (from MAST3R==0.0.1)
  Using cached trimesh-4.11.5-py3-none-any.whl.metadata (13 kB)
Collecting tensorboard (from MAST3R==0.0.1)
  Using cached tensorboard-2.20.0-py3-none-any.whl.metadata (1.8 kB)
Collecting pyglet (from MAST3R==0.0.1)
  Using cached pyglet-2.1.13-py3-none-any.whl.metadata (7.7 kB)
Collecting huggingface-hub>=0.22 (from
huggingface-hub[torch]>=0.22->MAST3R==0.0.1)
  Using cached huggingface_hub-1.8.0-py3-none-any.whl.metadata (13 kB)
Processing .\MASt3R-SLAM-API\thirdparty\mast3r\dust3r\croco\models\cuope
(from MAST3R==0.0.1)
  Preparing metadata (pyproject.toml) ... error
  error: subprocess-exited-with-error

  × Preparing metadata (pyproject.toml) did not run successfully.
  | exit code: 1
  └─> [20 lines of output]
      Traceback (most recent call last):
        File
"C:\Users\maste\miniconda3\Lib\site-packages\pip\_vendor\pyproject_hooks\
in_process\in_process.py", line 389, in <module>
          main()
          ~~~~^^
        File
"C:\Users\maste\miniconda3\Lib\site-packages\pip\_vendor\pyproject_hooks\
in_process\in_process.py", line 373, in main
          json_out["return_val"] = hook(**hook_input["kwargs"])
                                   ~~~~~^
        File
"C:\Users\maste\miniconda3\Lib\site-packages\pip\_vendor\pyproject_hooks\
in_process\in_process.py", line 175, in prepare_metadata_for_build_wheel
          return hook(metadata_directory, config_settings)
        File
"C:\Users\maste\miniconda3\Lib\site-packages\setuptools\build_meta.py",
line 378, in prepare_metadata_for_build_wheel
          self.run_setup()
          ~~~~~^
        File
"C:\Users\maste\miniconda3\Lib\site-packages\setuptools\build_meta.py",
line 518, in run_setup
          super().run_setup(setup_script=setup_script)
```


Brandon's Notes

```
~~~~~^~~~~~
File
"C:\Users\maste\miniconda3\Lib\site-packages\setuptools\build_meta.py",
line 317, in run_setup
    exec(code, locals())
    ~~~^~~~~~
File "<string>", line 5, in <module>
ModuleNotFoundError: No module named 'torch'
[end of output]
```

note: This error originates from a subprocess, and is likely not a problem with pip.

error: metadata-generation-failed

× Encountered error while generating package metadata.

└─> curope

note: This is an issue with the package mentioned above, not pip.

hint: See above for details.

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To check on environments:

```
conda info --envs
```

- Issue: Must install Microsoft Visual Studio
<https://visualstudio.microsoft.com/visual-cpp-build-tools/>
- Ensure the following are also installed for it
 - MSVC v143 (or latest)
 - Windows 10/11 SDK
 - C++ CMake tools for Windows
 -

```
pip install --no-build-isolation -e .\MASt3R-SLAM-API\thirdparty\mast3r
Obtaining
file:///C:/Users/maste/OneDrive/Documents/Engineering/CSCE/Csce_483/TeamBallers/MASt3R-SLAM-API/thirdparty/mast3r
Checking if build backend supports build_editable ... done
Preparing editable metadata (pyproject.toml) ... done
Collecting scikit-learn (from MAST3R==0.0.1)
Using cached scikit_learn-1.8.0-cp313-cp313-win_amd64.whl.metadata (11
kB)
Collecting roma (from MAST3R==0.0.1)
Using cached roma-1.5.6-py3-none-any.whl.metadata (5.5 kB)
```

Brandon's Notes

```
Collecting gradio (from MAST3R==0.0.1)
  Using cached gradio-6.10.0-py3-none-any.whl.metadata (17 kB)
Collecting matplotlib (from MAST3R==0.0.1)
  Using cached matplotlib-3.10.8-cp313-cp313-win_amd64.whl.metadata (52
kB)
Requirement already satisfied: tqdm in
C:\Users\maste\miniconda3\Lib\site-packages (from MAST3R==0.0.1) (4.67.3)
Collecting opencv-python (from MAST3R==0.0.1)
  Using cached opencv_python-4.13.0.92-cp37-abi3-win_amd64.whl.metadata
(20 kB)
Collecting scipy (from MAST3R==0.0.1)
  Using cached scipy-1.17.1-cp313-cp313-win_amd64.whl.metadata (60 kB)
Collecting einops (from MAST3R==0.0.1)
  Using cached einops-0.8.2-py3-none-any.whl.metadata (13 kB)
Collecting trimesh (from MAST3R==0.0.1)
  Using cached trimesh-4.11.5-py3-none-any.whl.metadata (13 kB)
Collecting tensorboard (from MAST3R==0.0.1)
  Using cached tensorboard-2.20.0-py3-none-any.whl.metadata (1.8 kB)
Collecting pyglet (from MAST3R==0.0.1)
  Using cached pyglet-2.1.13-py3-none-any.whl.metadata (7.7 kB)
Collecting huggingface-hub>=0.22 (from
huggingface-hub[torch]>=0.22->MAST3R==0.0.1)
  Using cached huggingface_hub-1.8.0-py3-none-any.whl.metadata (13 kB)
Processing .\MASt3R-SLAM-API\thirdparty\mast3r\dust3r\croco\models\cuope
(from MAST3R==0.0.1)
  Preparing metadata (pyproject.toml) ... error
  error: subprocess-exited-with-error

  × Preparing metadata (pyproject.toml) did not run successfully.
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  └─> [20 lines of output]
      Traceback (most recent call last):
        File
"C:\Users\maste\miniconda3\Lib\site-packages\pip\_vendor\pyproject_hooks\
in_process\in_process.py", line 389, in <module>
          main()
          ~~~~^^
        File
"C:\Users\maste\miniconda3\Lib\site-packages\pip\_vendor\pyproject_hooks\
in_process\in_process.py", line 373, in main
          json_out["return_val"] = hook(**hook_input["kwargs"])
                                   ~~~~~^
        File
"C:\Users\maste\miniconda3\Lib\site-packages\pip\_vendor\pyproject_hooks\
in_process\in_process.py", line 175, in prepare_metadata_for_build_wheel
          return hook(metadata_directory, config_settings)
```

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```
File
"C:\Users\maste\miniconda3\Lib\site-packages\setuptools\build_meta.py",
line 378, in prepare_metadata_for_build_wheel
    self.run_setup()
    ~~~~~^
File
"C:\Users\maste\miniconda3\Lib\site-packages\setuptools\build_meta.py",
line 518, in run_setup
    super().run_setup(setup_script=setup_script)
    ~~~~~^
File
"C:\Users\maste\miniconda3\Lib\site-packages\setuptools\build_meta.py",
line 317, in run_setup
    exec(code, locals())
    ~~~^
File "<string>", line 5, in <module>
ModuleNotFoundError: No module named 'torch'
[end of output]
```

note: This error originates from a subprocess, and is likely not a problem with pip.
error: metadata-generation-failed

× Encountered error while generating package metadata.
└> cuope

note: This is an issue with the package mentioned above, not pip.
hint: See above for details.

- Must check on torch with

```
python -c "import torch; print(torch.__version__)"
```

- If Torch missing:

```
python -m pip install --no-build-isolation -e
.\MASt3R-SLAM-API\thirdparty\mast3r
```

- Next Issue: need to install CUDA

https://developer.nvidia.com/cuda-downloads?target_os=Windows&target_arch=x86_64&target_version=11&target_type=exe_local

- Issue: Torch is not found. No matter how many times I uninstall and reinstall

```
python -c "import torch; print(torch.version.cuda)"
>>
None
```

Brandon's Notes

(mast3r-slam) PS

```
C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers> python -c "import torch; print(torch.cuda.is_available())"
```

False

- The current likely issue is now either: **CUDA 13.2 vs Torch 12.1 mismatch**, or **missing MSVC compiler tools**.
- May also need to create Python environment

Conclusion: I need CUDA 12.1 but that is impossible to install

- Now this command works

```
python -m pip install --no-build-isolation -e ".\MAST3R-SLAM-API\thirdparty\mast3r"
```

```
(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers> python -m pip install --no-build-isolation -e ".\MAST3R-SLAM-API\thirdparty\mast3r"
(from tensorboard->MAST3R==0.0.1) (2.4.0)
Requirement already satisfied: grpcio>=1.48.2 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from tensorboard->MAST3R==0.0.1) (1.80.0)
Requirement already satisfied: markdown>=2.6.8 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from tensorboard->MAST3R==0.0.1) (3.10.2)
Requirement already satisfied: protobuf!=4.24.0,>=3.19.6 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from tensorboard->MAST3R==0.0.1) (7.34.1)
Requirement already satisfied: setuptools>=41.0.0 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from tensorboard->MAST3R==0.0.1) (81.0.0)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from tensorboard->MAST3R==0.0.1) (0.7.2)
Requirement already satisfied: werkzeug>=1.0.1 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from tensorboard->MAST3R==0.0.1) (3.1.7)
Building wheels for collected packages: MAST3R, asmk, curope
  Building editable for MAST3R (pyproject.toml) ... done
  Created wheel for MAST3R: filename=mast3r-0.0.1-0.editable-py3-none-any.whl size=5162 sha256=d346d7325440f05590b98fadb0eeff13180ba6031d4644156eb5fc5897014d48
  Stored in directory: C:\Users\maste\AppData\Local\Temp\pip-ephem-wheel-cache-gxc132ml\wheels\9e\80\9b\0fd21635cac7ff979d2636b99976c5fc195cdbd388c9db5903
  Building wheel for asmk (pyproject.toml) ... done
  Created wheel for asmk: filename=asmk-0.1-py3-none-any.whl size=12768 sha256=9495ec3fcd79183f50935481a99a5785c4cf7884b1e361d165f8c7e76aa2e6c0
  Stored in directory: C:\Users\maste\AppData\Local\Temp\pip-ephem-wheel-cache-gxc132ml\wheels\ca\b9\21\fc01a3246b5cc9eedb1d1100c6f56e2364a3951efc7cb10843
  Building wheel for curope (pyproject.toml) ... done
  Created wheel for curope: filename=curope-0.0.0-py3-none-any.whl size=915 sha256=a2976a80c5387db08b578ec74b616043a0b4172e663f91abe26ffc543d2715b2
  Stored in directory: C:\Users\maste\AppData\Local\Temp\pip-ephem-wheel-cache-gxc132ml\wheels\98\c5\cb\77748c72509c0db3e08ea7ffac5f3036ae4d463443d3e0d79
Successfully built MAST3R asmk curope
Installing collected packages: curope, asmk, MAST3R
  Attempting uninstall: curope
    Found existing installation: curope 0.0.0
    Uninstalling curope-0.0.0:
      Successfully uninstalled curope-0.0.0
  Attempting uninstall: asmk
    Found existing installation: asmk 0.1
    Uninstalling asmk-0.1:
      Successfully uninstalled asmk-0.1
  Attempting uninstall: MAST3R
    Found existing installation: MAST3R 0.0.1
    Uninstalling MAST3R-0.0.1:
      Successfully uninstalled MAST3R-0.0.1
Successfully installed MAST3R-0.0.1 asmk-0.1 curope-0.0.0
```

Brandon's Notes

- now have this problem:

`python -m pip install --no-build-isolation "git+https://github.com/princeton-vl/lietorch.git"`

```
(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers> python -m pip install --no-build-isolation "git+https://github.com/princeton-vl/lietorch.git"
```

- https://setuptools.pypa.io/en/latest/userguide/package_discovery.html

If you don't want 'lietorch.src' to be distributed and are already explicitly excluding 'lietorch.src' via ``find_namespace_packages(...)/find_namespace`` or ``find_packages(...)/find``, you can try to use ``exclude_package_data``, or ``include_package_data=False`` in combination with a more fine grained ``package_data`` configuration.

You can read more about "package data files" on setuptools documentation page:

- <https://setuptools.pypa.io/en/latest/userguide/datafiles.html>

[^1]: For Python, any directory (with suitable naming) can be imported, even if it does not contain any ``.py`` files. On the other hand, currently there is no concept of package data directory, all directories are treated like packages.

!!

```
check.warn(importable)
creating build\lib.win-amd64-cpython-311\lietorch\extras
copying lietorch\extras\altcorr_kernel.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
copying lietorch\extras\corr_index_kernel.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
copying lietorch\extras\extras.cpp -> build\lib.win-amd64-cpython-311\lietorch\extras
copying lietorch\extras\se3_builder.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
copying lietorch\extras\se3_inplace_builder.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
copying lietorch\extras\se3_solver.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
creating build\lib.win-amd64-cpython-311\lietorch\src
copying lietorch\src\lietorch.cpp -> build\lib.win-amd64-cpython-311\lietorch\src
copying lietorch\src\lietorch_cpu.cpp -> build\lib.win-amd64-cpython-311\lietorch\src
copying lietorch\src\lietorch_gpu.cu -> build\lib.win-amd64-cpython-311\lietorch\src
running build_ext
error: [WinError 2] The system cannot find the file specified
[end of output]
```

note: This error originates from a subprocess, and is likely not a problem with pip.

ERROR: Failed building wheel for lietorch

Failed to build lietorch

error: failed-wheel-build-for-install

× Failed to build installable wheels for some pyproject.toml based projects

↳ lietorch

- Issue:
 - lietorch does not build on native Windows
 - wsl is not available on this machine right now
- options
 - Option 1 — Best path
 - Set up WSL2 / Ubuntu or use a Linux machine for the SLAM part.
 - Option 2 — Windows-only for now

Brandon's Notes

- Work on Gaussian Splatting separately on Windows using exported images/poses later, but not this MAST3R-SLAM runtime yet.

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What I've been able to do so far

- Install conda
- conda init powershell
- conda create -n mast3r-slam python=3.11 -y
- conda activate mast3r-slam
- conda install pytorch==2.5.1 torchvision==0.20.1 torchaudio==2.5.1 pytorch-cuda=12.1 -c pytorch -c nvidia
- python -m pip install --no-build-isolation -e ".\MASt3R-SLAM-API\thirdparty\mast3r"

Install WSL

```
wsl --install -d Ubuntu
```

- Got into WSL and done the following commands

```
mkdir -p ~/venvs
```

```
python3 -m venv ~/venvs/teamballers-wsl
```

```
source ~/venvs/teamballers-wsl/bin/activate
```

```
cd
```

```
/mnt/c/Users/maste/OneDrive/Documents/Engineering/CSCE/Csce_483/TeamBallers/MASt3R-SLAM-API
```

```
python -m pip install --upgrade pip setuptools wheel
```

```
python -m pip install torch torchvision torchaudio
```

```
python -m pip install numpy==1.26.4 einops natsort plyfile pillow tqdm evo opencv-python fastapi "uvicorn[standard]" python-multipart matplotlib scikit-image lpips
```

```
python -m pip install --no-build-isolation -e ./thirdparty/mast3r
```

```
python -m pip install --no-build-isolation "git+https://github.com/princeton-vl/lietorch.git"
```

```
python -m pip install --no-deps -e .
```

- Hit the same wall after "python -m pip install --no-build-isolation -e ./thirdparty/mast3r"

Kiet's Help

Issue: pip install git+<https://github.com/princeton-vl/lietorch.git>

- This is likely due to CUDA mismatch
- I have reached my API rate limit on Copilot so I can no longer get its help

PS

```
C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MASt3R-SLAM-API> set CUDA_HOME="C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v13.2"
```

```
(mast3r-slam) PS
```

```
C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MASt3R-SLAM-API> echo $env:CUDA_HOME
```

```
C:\Program Files\NVIDIA GPU Computing Toolkit\CUDA\v12.1
```

- Decision: Skip lietorch and move on to the other steps in Kiet's Guide
- Download for this is extremely slow:

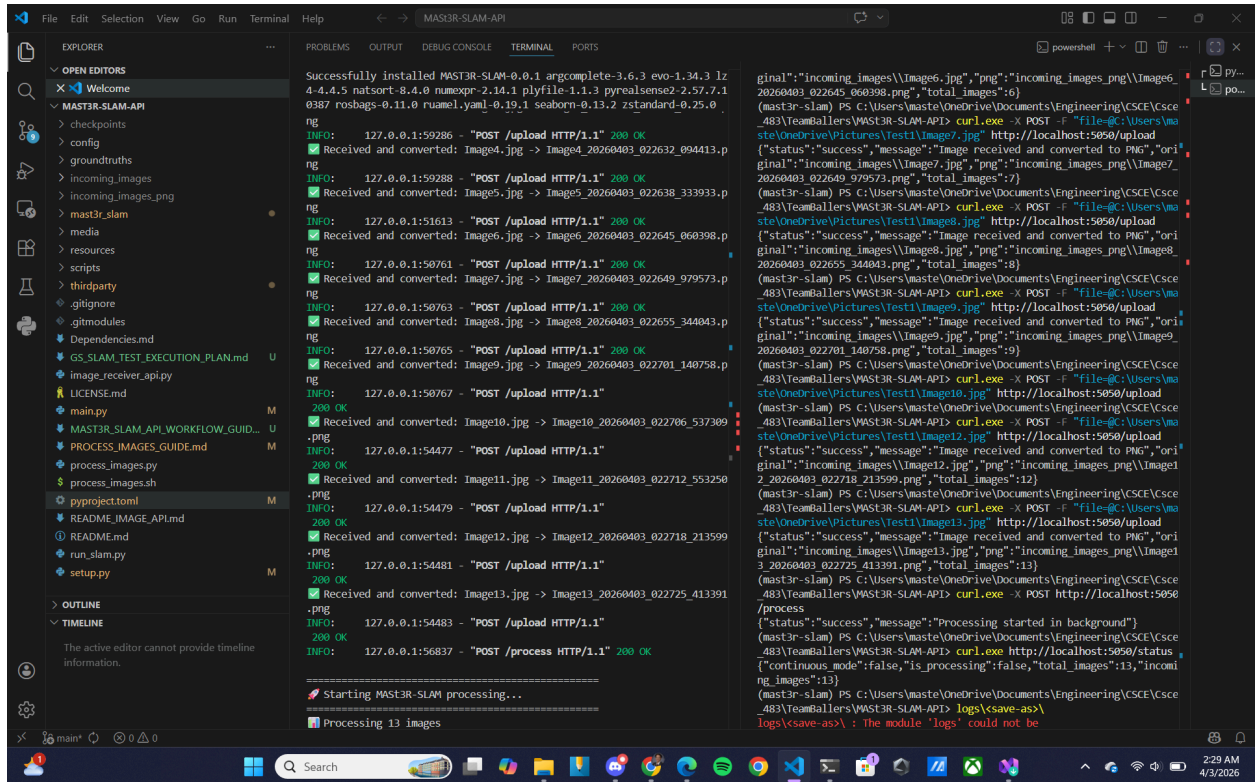
Brandon's Notes

- powershell -Command "Invoke-WebRequest -Uri 'https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric.pth' -OutFile 'checkpoints/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric.pth'"
- Invoke-WebRequest : Received an unexpected EOF or 0 bytes from the transport stream.
- At line:1 char:1
- + Invoke-WebRequest -Uri '<https://download.europe.naverlabs.com/Compute ...>'
- +
~~~~~  
~~~~~
- + CategoryInfo : NotSpecified: (:) [Invoke-WebRequest], IOException
- + FullyQualifiedErrorId : [System.IO.IOException](#),Microsoft.PowerShell.Commands.InvokeWebRequestCommand

```
pip install -e thirdparty/in3d
```

- ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
opencv-python 4.13.0.92 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.
Successfully installed attrs-26.1.0 charset-normalizer-3.4.7 colorlog-6.10.1 glcontext-3.0.0 glfw-2.10.0 imgui-2.0.0 in3d-0.0.1 jsonschema-4.26.0 jsonschema-specifications-2025.9.1 lxml-6.0.2 manifold3d-3.4.1 mapbox_earcut-2.0.0 moderngl-5.12.0 moderngl-window-2.4.6 msgpack-1.1.2 multipledispatch-1.0.0 numpy-1.26.4 pycollada-0.9.3 pyglm-2.8.3 pyrr-0.10.3 referencing-0.37.0 rpds-py-0.30.0 rtree-1.4.1 shapely-2.1.2 svg.path-7.0 vhacd-0.0.10 xxhash-3.6.0
- Manually took photos on camera from my own laptop and uploaded them to a folder. Then manually ran process_images.py in a split terminal and uploaded each photo one at a time into a copied folder and into the server.

Brandon's Notes



```
File Edit Selection View Go Run Terminal Help
MAST3R-SLAM-API

EXPLORER
OPEN EDITORS
Welcome
MAST3R-SLAM-API
  checkpoints
  config
  groundtruths
  incoming_images
  incoming_images.png
  mast3r_slam
  media
  resources
  scripts
  thirdparty
  .gitignore
  .gitmodules
  Dependencies.md
  GS_SLAM_TEST_EXECUTION_PLAN.md
  image_receiver_api.py
  LICENSE.md
  main.py
  MAST3R_SLAM_API_WORKFLOW_GUIDE...
  PROCESS_IMAGES_GUIDE.md
  process_images.py
  process_images.sh
  pyproject.toml
  README_IMAGE_API.md
  README.md
  run_slam.py
  setup.py

OUTLINE
TIMELINE
The active editor cannot provide timeline information.

TERMINAL
Successfully installed MAST3R-SLAM-0.0.1 argcomplete-3.6.3 evo-1.34.3 lz4-4.4.4.5 natsort-8.4.0 numexpr-2.14.1 plyfile-1.1.3 pyrealsense2-2.57.7.1 rosbags-0.11.0 ruamel.yaml-0.19.1 seaborn-0.13.2 zstandard-0.25.0
INFO: 127.0.0.1:59286 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Image4.jpg -> Image4_20260403_022632_094413.p
INFO: 127.0.0.1:59288 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Images.jpg -> Image5_20260403_022638_333933.p
INFO: 127.0.0.1:51613 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Image6.jpg -> Image6_20260403_022645_060398.p
INFO: 127.0.0.1:50761 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Image7.jpg -> Image7_20260403_022649_979573.p
INFO: 127.0.0.1:50763 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Image8.jpg -> Image8_20260403_022655_344043.p
INFO: 127.0.0.1:50765 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Image9.jpg -> Image9_20260403_022701_140758.p
INFO: 127.0.0.1:50767 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Image10.jpg -> Image10_20260403_022706_537309.p
INFO: 127.0.0.1:54477 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Image11.jpg -> Image11_20260403_022712_553250.p
INFO: 127.0.0.1:54479 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Image12.jpg -> Image12_20260403_022718_213599.p
INFO: 127.0.0.1:54481 - "POST /upload HTTP/1.1" 200 OK
Received and converted: Image13.jpg -> Image13_20260403_022725_413391.p
INFO: 127.0.0.1:54483 - "POST /upload HTTP/1.1" 200 OK
INFO: 127.0.0.1:56837 - "POST /process HTTP/1.1" 200 OK

Starting MAST3R-SLAM processing...
Processing 13 images

(gist) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MAST3R-SLAM-API> python process_images.py --input "C:\Users\maste\OneDrive\Pictures\Test1" --save-as test1 --no-viz

Processing failed!

(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MAST3R-SLAM-API> python process_images.py --input incoming_images --save-as first_run --no-viz

MAST3R-SLAM Image Processing Pipeline

Input directory: incoming_images
Output directory: incoming_images.png
Config file: config/base.yaml
Results name: first_run

Found 26 images
Converting to PNG format...
Converting: 100% 26/26 [00:04:00:00, 5.93it/s]
Successfully converted 26 images

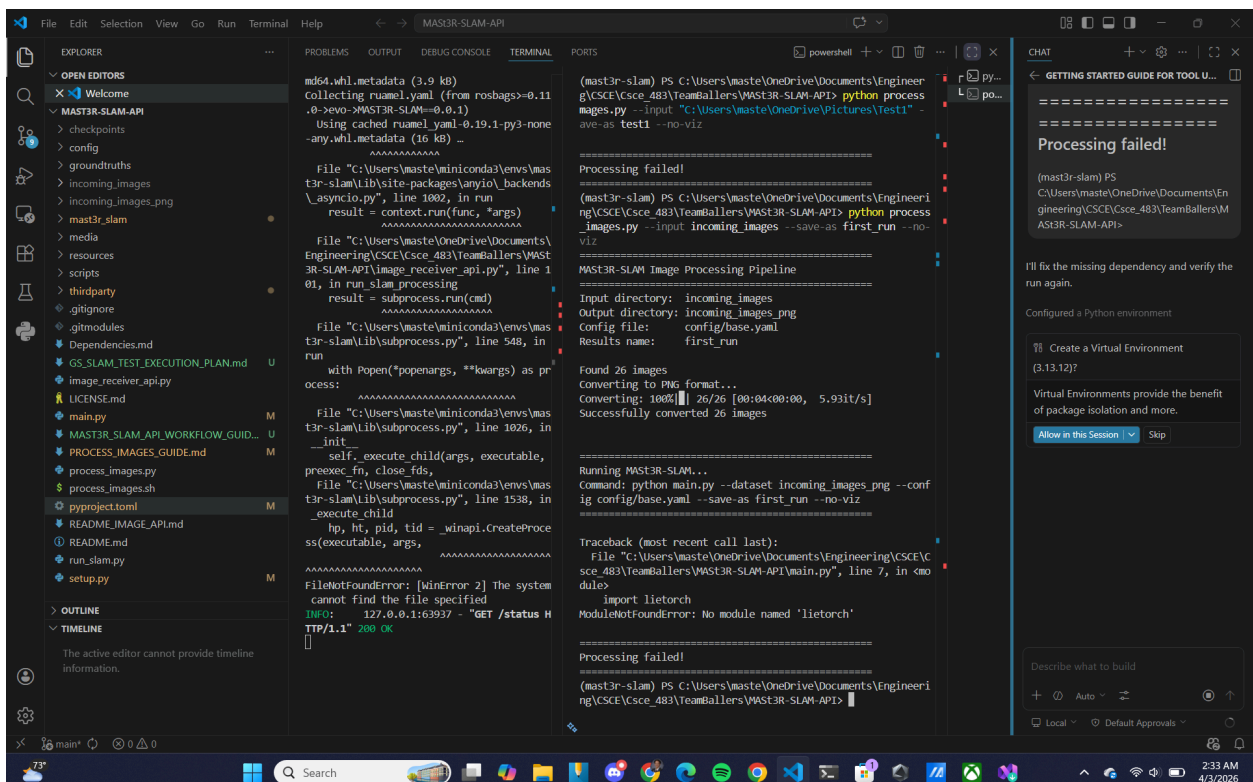
Running MAST3R-SLAM...
Command: python main.py --dataset incoming_images.png --config config/base.yaml --save-as first_run --no-viz

Traceback (most recent call last):
  File "C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MAST3R-SLAM-API\main.py", line 7, in <module>
    import lietorch
ModuleNotFoundError: No module named 'lietorch'

Processing failed!

(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MAST3R-SLAM-API> logs<save-as> logs<save-as> : The module 'logs' could not be
```

- In order to process them as png files, I need lietorch to work but I skipped lietorch
- It claims to fail but still show up on the files.



```
File Edit Selection View Go Run Terminal Help
MAST3R-SLAM-API

EXPLORER
OPEN EDITORS
Welcome
MAST3R-SLAM-API
  checkpoints
  config
  groundtruths
  incoming_images
  incoming_images.png
  mast3r_slam
  media
  resources
  scripts
  thirdparty
  .gitignore
  .gitmodules
  Dependencies.md
  GS_SLAM_TEST_EXECUTION_PLAN.md
  image_receiver_api.py
  LICENSE.md
  main.py
  MAST3R_SLAM_API_WORKFLOW_GUIDE...
  PROCESS_IMAGES_GUIDE.md
  process_images.py
  process_images.sh
  pyproject.toml
  README_IMAGE_API.md
  README.md
  run_slam.py
  setup.py

OUTLINE
TIMELINE
The active editor cannot provide timeline information.

TERMINAL
md64.whl.metadata (3.9 kB)
Collecting ruamel.yaml (from rosbags==0.11.0)
Using cached ruamel.yaml-0.19.1-py3-none-any.whl.metadata (16 kB)
File "C:\Users\maste\miniconda3\envs\mas...
t3r-slam\lib\site-packages\anyio\backends\
_asyncio.py", line 1082, in run
    result = context.run(func, *args)
File "C:\Users\maste\OneDrive\Documents\
Engineering\CSCE_483\TeamBallers\MAST3R-SLAM-API\image_receiver_api.py", line 1
01, in run_slam_processing
    result = subprocess.run(cmd)
File "C:\Users\maste\miniconda3\envs\mas...
t3r-slam\lib\subprocess.py", line 548, in
run
    with Popen(*popenargs, **kwargs) as p
rocess:
File "C:\Users\maste\miniconda3\envs\mas...
t3r-slam\lib\subprocess.py", line 1826, in
__init__
    self._execute_child(args, executable,
preexec_fn, close_fds,
File "C:\Users\maste\miniconda3\envs\mas...
t3r-slam\lib\subprocess.py", line 1538, in
_execute_child
    hp, ht, pid, tid = _winapi.CreateProce
ss(executable, args,
FileNotFoundError: [WinError 2] The system
cannot find the file specified
INFO: 127.0.0.1:63937 - "GET /status H
TTP/1.1" 200 OK

(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MAST3R-SLAM-API> python process_images.py --input "C:\Users\maste\OneDrive\Pictures\Test1" --save-as test1 --no-viz

Processing failed!

(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MAST3R-SLAM-API> python process_images.py --input incoming_images --save-as first_run --no-viz

MAST3R-SLAM Image Processing Pipeline

Input directory: incoming_images
Output directory: incoming_images.png
Config file: config/base.yaml
Results name: first_run

Found 26 images
Converting to PNG format...
Converting: 100% 26/26 [00:04:00:00, 5.93it/s]
Successfully converted 26 images

Running MAST3R-SLAM...
Command: python main.py --dataset incoming_images.png --config config/base.yaml --save-as first_run --no-viz

Traceback (most recent call last):
  File "C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MAST3R-SLAM-API\main.py", line 7, in <module>
    import lietorch
ModuleNotFoundError: No module named 'lietorch'

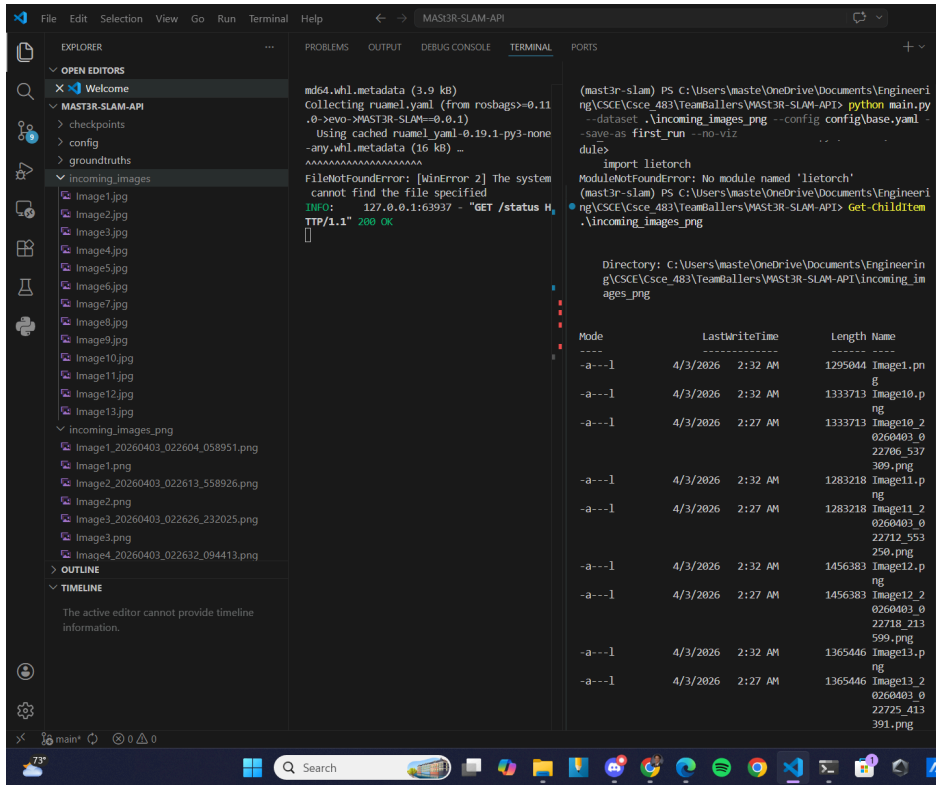
Processing failed!

(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MAST3R-SLAM-API> logs<save-as> logs<save-as> : The module 'logs' could not be

CHAT
GETTING STARTED GUIDE FOR TOOL U...
Processing failed!
(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers\MAST3R-SLAM-API>
I'll fix the missing dependency and verify the run again.
Configured a Python environment
Create a Virtual Environment (3.13.12)?
Virtual Environments provide the benefit of package isolation and more.
Allow in this Session Skip
```


Brandon's Notes

- Proves I can upload photos without lietorch



```
(mast3r-slam) PS C:\Users\waste\OneDrive\Documents\Engineering\CSCE\csce_483\TeamBallers\WASTE3R-SLAM-API> python main.py --dataset .\incoming_images.png --config config\base.yaml --save-as first_run --no-viz ...
import lietorch
ModuleNotFoundError: No module named 'lietorch'
(mast3r-slam) PS C:\Users\waste\OneDrive\Documents\Engineering\CSCE\csce_483\TeamBallers\WASTE3R-SLAM-API> Get-Childitem .\incoming_images.png

Mode                LastWriteTime         Length Name
----                -
-a---1            4/3/2026   2:32 AM          1295044 Image1.png
-a---1            4/3/2026   2:32 AM          1333713 Image10.png
-a---1            4/3/2026   2:27 AM          1333713 Image10_2_0260403_022706_537309.png
-a---1            4/3/2026   2:32 AM          1283218 Image11.png
-a---1            4/3/2026   2:27 AM          1283218 Image11_2_0260403_022712_553250.png
-a---1            4/3/2026   2:32 AM          1456383 Image12.png
-a---1            4/3/2026   2:27 AM          1456383 Image12_2_0260403_022718_213599.png
-a---1            4/3/2026   2:32 AM          1365446 Image13.png
-a---1            4/3/2026   2:27 AM          1365446 Image13_2_0260403_022725_413391.png
```

What is blocking the SLAM run

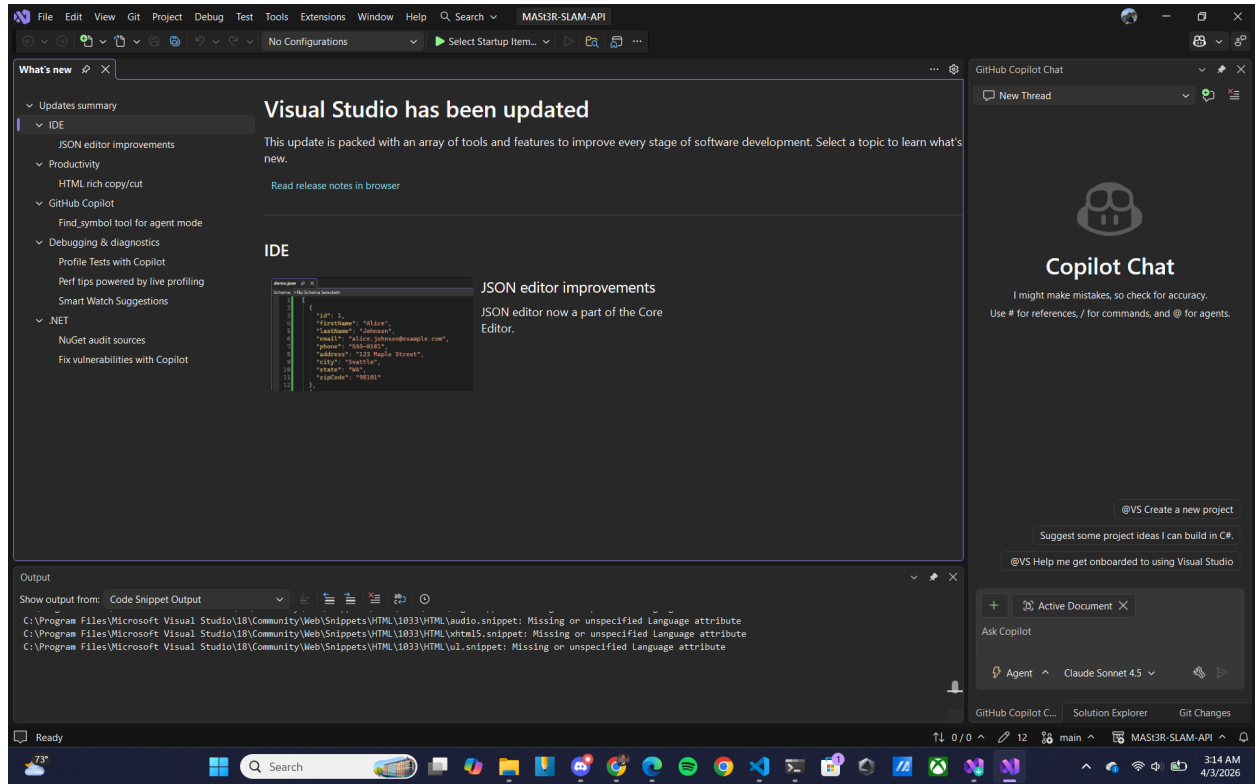
I verified the environment and found:

- GPU is detected: `nvidia-smi` shows your RTX 5070
- PyTorch is now GPU-enabled in `mast3r-slam`:
 - torch 2.5.1
 - CUDA 12.4
 - `torch.cuda.is_available() = True`

But installing lietorch still fails because:

1. `cl.exe` is missing
→ Microsoft C++ Build Tools are not installed
 2. CUDA toolkit mismatch
→ your system toolkit is CUDA 13.2, while PyTorch was built with CUDA 12.4
- It's strange that it claims that the main issue is that I don't have the C++ build tool on Microsoft Visual Studio but I clearly have it here.

Brandon's Notes



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What to show at weekly meeting:

conda activate mast3r-slam //if not in environment already

conda install pytorch torchvision torchaudio pytorch-cuda=13.2 -c pytorch -c nvidia -y

python -m pip install torch torchvision torchaudio //works

pip install -e thirdparty/mast3r //error but still says successfully installed

Brandon's Notes

pip install -e thirdparty/in3d //error but still says successfully installed

```
(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CsCE\Csce_483\TeamBallers\MAST3R-SLAM-API> pip install -e thirdparty/in3d
sh[easy]->in3d==0.0.1) (3.11)
Requirement already satisfied: h11>=0.16 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from httpcor
e==1.*->httpx->trimesh[easy]->in3d==0.0.1) (0.16.0)
Requirement already satisfied: typing_extensions>=4.5 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages
(from anyio->httpx->trimesh[easy]->in3d==0.0.1) (4.15.0)
Requirement already satisfied: attrs>=22.2.0 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from jso
nschema->trimesh[easy]->in3d==0.0.1) (26.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\
site-packages (from jsonschema->trimesh[easy]->in3d==0.0.1) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (fr
om jsonschema->trimesh[easy]->in3d==0.0.1) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from j
sonschema->trimesh[easy]->in3d==0.0.1) (0.30.0)
Using cached numpy-1.26.4-cp311-cp311-win_amd64.whl (15.8 MB)
Building wheels for collected packages: in3d, imgui
  Building editable for in3d (pyproject.toml) ... done
  Created wheel for in3d: filename=in3d-0.0.1-0.editable-py3-none-any.whl size=2841 sha256=6c0c1f0d612f8bb871828c132a9
271f1625e0b6531826152ba8d2393bbca596d
  Stored in directory: C:\Users\maste\AppData\Local\Temp\pip-ephem-wheel-cache-w0187zaj\wheels\4d\bc\ae\02ff2ef46493447
4329484decaa65f87da6eaeca77595a166cb
  Building wheel for imgui (pyproject.toml) ... done
  Created wheel for imgui: filename=imgui-2.0.0-cp311-cp311-win_amd64.whl size=1624882 sha256=ec814acecfbbfa897e85bdf2
e30ee2c50e9a20ec361d56adf44ff8d2e4d012d6
  Stored in directory: C:\Users\maste\AppData\Local\Temp\pip-ephem-wheel-cache-w0187zaj\wheels\ae\b0\49\65b3f827a20469
8b8b2517768d3cd550a9e7863ced285e86c
Successfully built in3d imgui
Installing collected packages: imgui, numpy, in3d
  Attempting uninstall: imgui
    Found existing installation: imgui 2.0.0
    Uninstalling imgui-2.0.0:
      Successfully uninstalled imgui-2.0.0
  Attempting uninstall: numpy
    Found existing installation: numpy 2.3.1
    Uninstalling numpy-2.3.1:
      Successfully uninstalled numpy-2.3.1
  Attempting uninstall: in3d
    Found existing installation: in3d 0.0.1
    Uninstalling in3d-0.0.1:
      Successfully uninstalled in3d-0.0.1
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behavi
our is the source of the following dependency conflicts.
opencv-python 4.13.0.92 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.
Successfully installed imgui-2.0.0 in3d-0.0.1 numpy-1.26.4
(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CsCE\Csce_483\TeamBallers\MAST3R-SLAM-API>
```

pip install --no-build-isolation -e .

```
Obtaining file:///C:/Users/maste/OneDrive/Documents/Engineering/CsCE/Csce_483/TeamBallers/MAST3R-SLAM-API
Checking if build backend supports build_editable ... done
Preparing editable metadata (pyproject.toml) ... done
Requirement already satisfied: numpy==1.26.4 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from MAS
T3R-SLAM==0.0.1) (1.26.4)
Requirement already satisfied: einops in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from MAST3R-SLA
M==0.0.1) (0.8.2)
Requirement already satisfied: pyrealsense2 in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from MAST
3R-SLAM==0.0.1) (2.57.7.10387)
Requirement already satisfied: evo in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from MAST3R-SLAM==
0.0.1) (1.34.3)
Requirement already satisfied: natsort in C:\Users\maste\miniconda3\envs\mast3r-slam\Lib\site-packages (from MAST3R-SL
AM==0.0.1) (8.4.0)
INFO: pip is looking at multiple versions of mast3r-slam to determine which version is compatible with other requireme
nts. This could take a while.
ERROR: Could not find a version that satisfies the requirement torchcodec==0.1 (from mast3r-slam) (from versions: 0.0.
0.dev0, 0.7.0, 0.8.0, 0.8.1, 0.9.0, 0.9.1, 0.10.0, 0.11.0)
ERROR: No matching distribution found for torchcodec==0.1
(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CsCE\Csce_483\TeamBallers\MAST3R-SLAM-API>
```

Brandon's Notes

Install lietorch (should compile cleanly with CUDA)

```
python -m pip install --no-build-isolation "git+https://github.com/princeton-vl/lietorch.git"
```

```
(mast3r-slam) PS C:\Users\maste\OneDrive\Documents\Engineering\CSCE\Csce_483\TeamBallers> python -m pip install --no-build-isolation "git+https://github.com/princeton-vl/lietorch.git"
```

```
- https://setuptools.pypa.io/en/latest/userguide/package_discovery.html
```

```
If you don't want 'lietorch.src' to be distributed and are already explicitly excluding 'lietorch.src' via `find_namespace_packages(...)/find_namespace` or `find_packages(...)/find`, you can try to use `exclude_package_data`, or `include_package_data=False` in combination with a more fine grained `package_data` configuration.
```

```
You can read more about "package data files" on setuptools documentation page:
```

```
- https://setuptools.pypa.io/en/latest/userguide/datafiles.html
```

```
[^1]: For Python, any directory (with suitable naming) can be imported, even if it does not contain any `.py` files.
```

```
On the other hand, currently there is no concept of package data directory, all directories are treated like packages.
```

```
*****
```

```
!!
```

```
check.warn(importable)
```

```
creating build\lib.win-amd64-cpython-311\lietorch\extras
```

```
copying lietorch\extras\altcorr_kernel.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
```

```
copying lietorch\extras\corr_index_kernel.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
```

```
copying lietorch\extras\extras.cpp -> build\lib.win-amd64-cpython-311\lietorch\extras
```

```
copying lietorch\extras\se3_builder.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
```

```
copying lietorch\extras\se3_inplace_builder.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
```

```
copying lietorch\extras\se3_solver.cu -> build\lib.win-amd64-cpython-311\lietorch\extras
```

```
creating build\lib.win-amd64-cpython-311\lietorch\src
```

```
copying lietorch\src\lietorch.cpp -> build\lib.win-amd64-cpython-311\lietorch\src
```

```
copying lietorch\src\lietorch_cpu.cpp -> build\lib.win-amd64-cpython-311\lietorch\src
```

```
copying lietorch\src\lietorch_gpu.cu -> build\lib.win-amd64-cpython-311\lietorch\src
```

```
running build_ext
```

```
error: [WinError 2] The system cannot find the file specified
```

```
[end of output]
```

```
note: This error originates from a subprocess, and is likely not a problem with pip.
```

```
ERROR: Failed building wheel for lietorch
```

```
Failed to build lietorch
```

```
error: failed-wheel-build-for-install
```

```
x Failed to build installable wheels for some pyproject.toml based projects
```

```
↳ lietorch
```

Verification steps:

```
nvcc --version
```

```
nvcc: NVIDIA (R) Cuda compiler driver
```

```
Copyright (c) 2005-2026 NVIDIA Corporation
```

```
Built on Mon_Mar__2_21:54:11_Pacific_Standard_Time_2026
```

```
Cuda compilation tools, release 13.2, V13.2.51
```

```
Build cuda_13.2.r13.2/compiler.37434383_0
```

```
python -c "import torch; print(torch.__version__)" //2.5.1+cpu
```

```
python -c "from torch.utils.cpp_extension import CUDA_HOME; print(CUDA_HOME)" //None
```

```
python -c "import torch; print(torch.cuda.is_available())" //False
```

Brandon's Notes

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- Redid everything outside of my OneDrive folder

Install in3d without conflict with numpy version

```
python -m pip install "numpy==1.26.4" "opencv-python==4.11.0.86"
python -m pip install -e thirdparty/in3d
```

- Reinstalled Cuda but with 12.4 to make sure there is no confusion on what version
- Because of that I need an older version of Microsoft Visual Studio in order for this command to work

- `pip install --no-build-isolation -e .`
-

- Instead I used these commands

```
python -m pip install --no-build-isolation -e thirdparty/lietorch
python -m pip install pyrealsense2 evo natsort plyfile
python -m pip install --no-build-isolation -e . --no-deps
```

- Instead of this command:

`wget`

```
https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric.pth -P checkpoints/
```

- Use this:

`New-Item -ItemType Directory -Force -Path checkpoints | Out-Null`

`Invoke-WebRequest`

`"https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric.pth" -OutFile`

`"checkpoints/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric.pth"`

- But takes forever → nothing happened in the end
- Now do this

`Invoke-WebRequest`

`"https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_trainingfree.pth" -OutFile`

`"checkpoints/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_trainingfree.pth"`

- Instead of this

`wget`

```
https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_trainingfree.pth -P checkpoints/
```

- Use this

Brandon's Notes

Invoke-WebRequest

```
"https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_codebook.pkl" -OutFile
```

```
"checkpoints/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_codebook.pkl"
```

- Instead of this

```
wget
```

```
https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_codebook.pkl -P checkpoints/
```

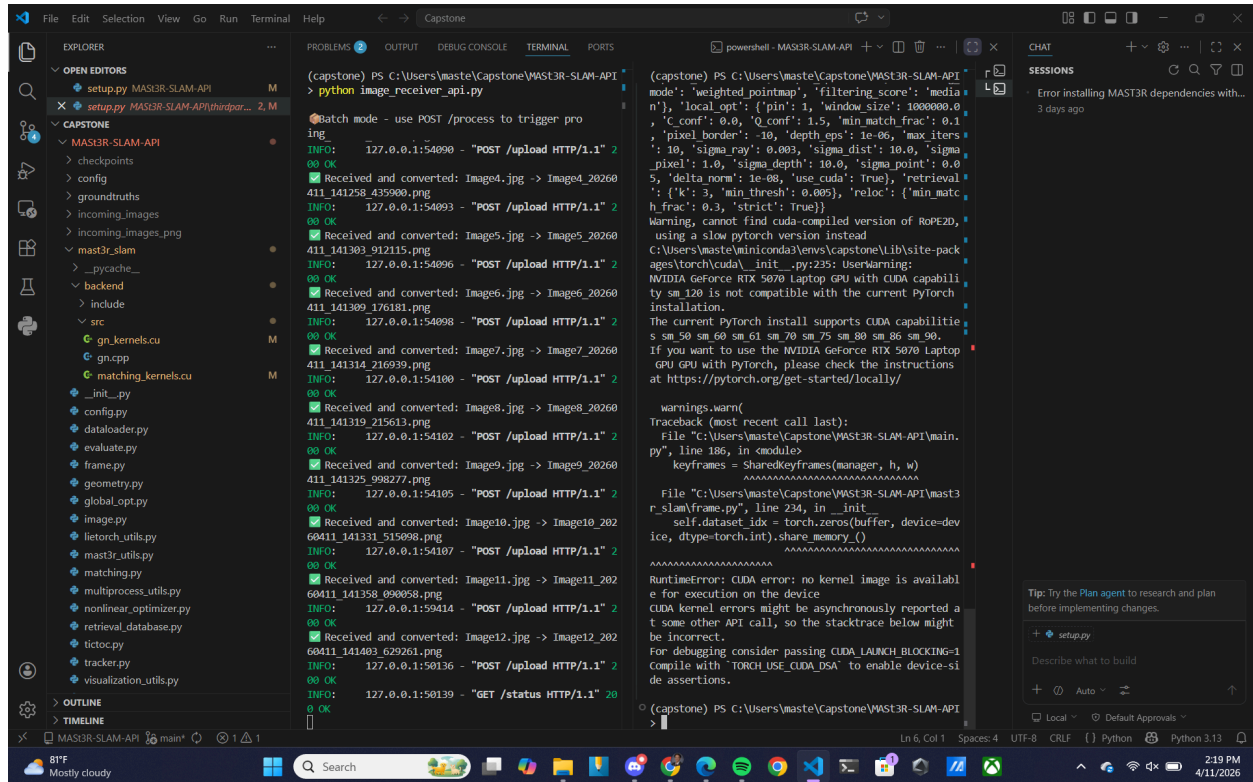
-

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- Checking on process_images.py

```
python process_images.py --input incoming_images --save-as first_run --no-viz
```


Brandon's Notes



- Command can't seem to give this error but still converts the images to a png
- The issue seems to stem from the version of CUDA I have installed

Used this command:

```
python -m pip uninstall -y torch torchvision torchaudio
python -m pip install --upgrade pip
python -m pip install torch torchvision torchaudio --index-url
https://download.pytorch.org/whl/cu128
```

- Fixed but now getting this

```
python main.py --dataset .\incoming_images_png --config config\base.yaml --save-as first_run
--no-viz
```

Traceback (most recent call last):

File "C:\Users\maste\Capstone\MASt3R-SLAM-API\main.py", line 7, in <module>

import lietorch

File "C:\Users\maste\Capstone\MASt3R-SLAM-API\thirdparty\lietorch\lietorch__init__.py", line 2, in <module>

from .groups import LieGroupParameter, SO3, RxSO3, SE3, Sim3, cat, stack

File "C:\Users\maste\Capstone\MASt3R-SLAM-API\thirdparty\lietorch\lietorch\groups.py", line 5, in <module>

from .group_ops import Exp, Log, Inv, Mul, Adj, AdjT, Jinv, Act3, Act4, ToVec, FromVec

File "C:\Users\maste\Capstone\MASt3R-SLAM-API\thirdparty\lietorch\lietorch\group_ops.py", line 1, in <module>

import lietorch_backends

Brandon's Notes

ImportError: DLL load failed while importing lietorch_backends: The specified procedure could not be found.

- The main blocker is no longer Python or PyTorch itself. The blocker is that the lietorch CUDA extension does not rebuild cleanly on the current Windows setup
- Lietorch successfully installed using Ubuntu
 - Username: maste@Newgate
 - Password: Masterle
- Now it gives errors on missing modules like "mast3r_slam_backends"

```
(.venv) maste@Newgate:/mnt/c/Users/maste/Capstone/MAS
t3R-SLAM-API$ python main.py --dataset ./incoming_ima
ges_png --config config/base.yaml --save-as first_run
--no-viz
Warning, cannot find cuda-compiled version of RoPE2D,
using a slow pytorch version instead
Traceback (most recent call last):
  File "/mnt/c/Users/maste/Capstone/MASt3R-SLAM-API/m
ain.py", line 11, in <module>
    from mast3r_slam.global_opt import FactorGraph
  File "/mnt/c/Users/maste/Capstone/MASt3R-SLAM-API/m
ast3r_slam/global_opt.py", line 4, in <module>
    from mast3r_slam.frame import SharedKeyframes
  File "/mnt/c/Users/maste/Capstone/MASt3R-SLAM-API/m
ast3r_slam/frame.py", line 6, in <module>
    from mast3r_slam.mast3r_utils import resize_img
  File "/mnt/c/Users/maste/Capstone/MASt3R-SLAM-API/m
ast3r_slam/mast3r_utils.py", line 11, in <module>
    import mast3r_slam.matching as matching
  File "/mnt/c/Users/maste/Capstone/MASt3R-SLAM-API/m
ast3r_slam/matching.py", line 5, in <module>
    import mast3r_slam_backends
ModuleNotFoundError: No module named 'mast3r_slam_bac
kends'
(.venv) maste@Newgate:/mnt/c/Users/maste/Capstone/MAS
t3R-SLAM-API$ export CUDA_HOME=/usr/local/cuda
(.venv) maste@Newgate:/mnt/c/Users/maste/Capstone/MAS
t3R-SLAM-API$ export PATH=$CUDA_HOME/bin:$PATH
(.venv) maste@Newgate:/mnt/c/Users/maste/Capstone/MAS
t3R-SLAM-API$ export TORCH_CUDA_ARCH_LIST="9.0+PTX"
(.venv) maste@Newgate:/mnt/c/Users/maste/Capstone/MAS
t3R-SLAM-API$ python setup.py build_ext --inplace
/mnt/c/Users/maste/Capstone/MASt3R-SLAM-API/.venv/lib
/python3.12/site-packages/wheel/bdist_wheel.py:4: Fut
ureWarning: The 'wheel' package is no longer the cano
nical location of the 'bdist_wheel' command, and will
be removed in a future release. Please update to set
uptools v70.1 or later which contains an integrated v
ersion of this command.
  warn(
running build_ext
building 'mast3r_slam_backends' extension
```

How to open on WSL:

```
python3 -m venv .venv
source .venv/bin/activate
```

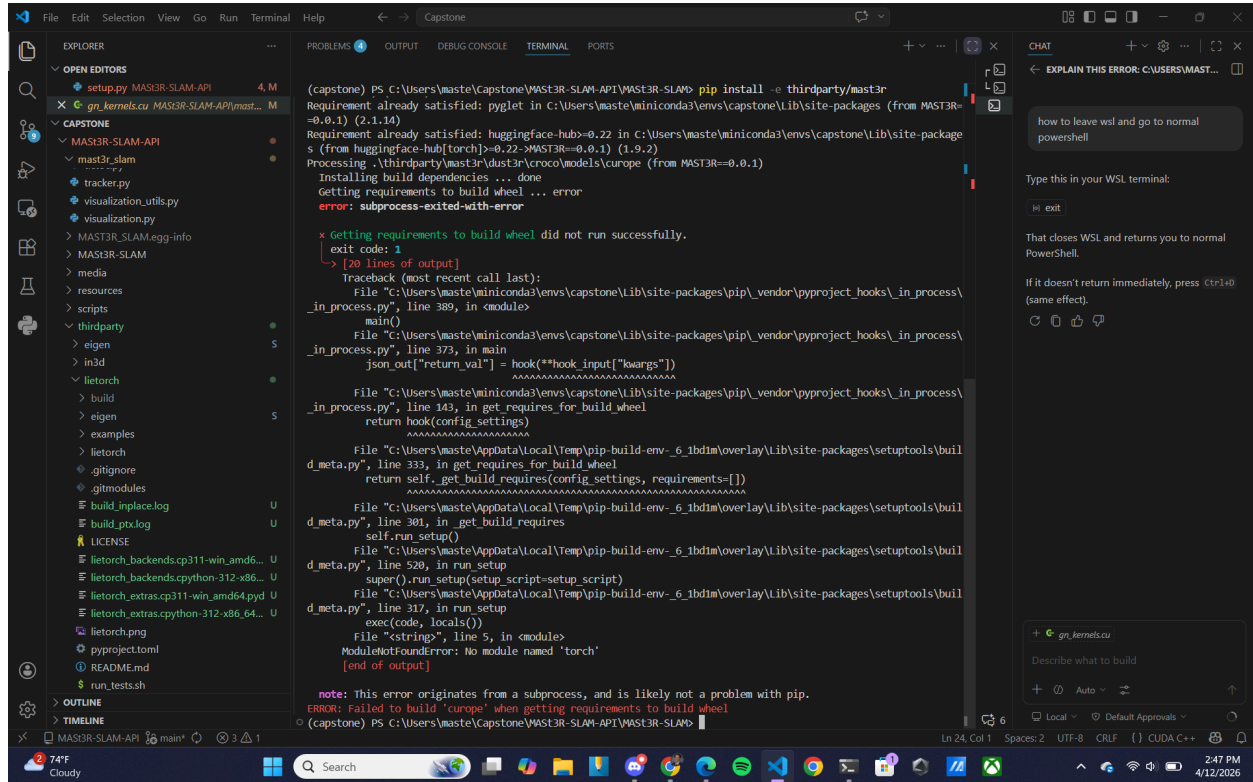

Brandon's Notes

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Redo with this README on capstone normal environment:

<https://github.com/TeamBallers/MASt3R-SLAM-API?tab=readme-ov-file>

When trying to install mast3r I get this old issue



```
(capstone) PS C:\Users\maste\capstone\MASt3R-SLAM-API\MASt3R-SLAM> pip install -e thirdparty/mast3r
Requirement already satisfied: pyglet in C:\Users\maste\miniconda3\envs\capstone\Lib\site-packages (from MASt3R==0.0.1) (2.1.14)
Requirement already satisfied: huggingface-hub>=0.22 in C:\Users\maste\miniconda3\envs\capstone\Lib\site-packages (from huggingface-hub[torch]>=0.22->MASt3R==0.0.1) (1.9.2)
Processing .\thirdparty\mast3r\dust3r\torch\models\europa (from MASt3R==0.0.1)
Installing build dependencies ... done
Getting requirements to build wheel ... error
error: subprocess-exited-with-error

x Getting requirements to build wheel did not run successfully.
  exit code: 1
  [20 lines of output]
  Traceback (most recent call last):
    File "C:\Users\maste\miniconda3\envs\capstone\Lib\site-packages\pip\_vendor\pyproject_hooks\_in_process\_in_process.py", line 389, in <module>
      main()
    File "C:\Users\maste\miniconda3\envs\capstone\Lib\site-packages\pip\_vendor\pyproject_hooks\_in_process\_in_process.py", line 373, in main
      json_out["return_val"] = hook(**hook_input["kwargs"])
    File "C:\Users\maste\miniconda3\envs\capstone\Lib\site-packages\pip\_vendor\pyproject_hooks\_in_process\_in_process.py", line 143, in get_requires_for_build_wheel
      return hook(config_settings)
    File "C:\Users\maste\AppData\Local\Temp\pip-build-env-6_1bd1m\overlay\Lib\site-packages\setuptools\build_dmeta.py", line 333, in get_requires_for_build_wheel
      return self.get_build_requires(config_settings, requirements=[])
    File "C:\Users\maste\AppData\Local\Temp\pip-build-env-6_1bd1m\overlay\Lib\site-packages\setuptools\build_dmeta.py", line 301, in get_build_requires
      self.run_setup()
    File "C:\Users\maste\AppData\Local\Temp\pip-build-env-6_1bd1m\overlay\Lib\site-packages\setuptools\build_dmeta.py", line 520, in run_setup
      super().run_setup(setup_script=setup_script)
    File "C:\Users\maste\AppData\Local\Temp\pip-build-env-6_1bd1m\overlay\Lib\site-packages\setuptools\build_dmeta.py", line 317, in run_setup
      exec(code, locals())
    File "<string>", line 5, in <module>
      ModuleNotFoundError: No module named 'torch'
  [end of output]

note: This error originates from a subprocess, and is likely not a problem with pip.
ERROR: Failed to build 'europa' when getting requirements to build wheel
(capstone) PS C:\Users\maste\capstone\MASt3R-SLAM-API\MASt3R-SLAM>
```

These 3 work now:

```
pip install -e thirdparty/mast3r # use --no-build-isolation if torch is not found
```

```
pip install -e thirdparty/in3d
```

```
pip install --no-build-isolation -e .
```

Now moving on to these commands:

```
mkdir -p checkpoints/
```

```
wget
```

```
https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric.pth -P checkpoints/
```

```
wget
```

```
https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_trainingfree.pth -P checkpoints/
```

```
wget
```

```
https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_codebook.pkl -P checkpoints/
```

Brandon's Notes

- Need to do this instead:
 - Invoke-WebRequest -Uri
"https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric.pth" -OutFile
"checkpoints\MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric.pth"
 - Only this one takes forever
 - Invoke-WebRequest -Uri
"https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_trainingfree.pth" -OutFile
"checkpoints\MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_trainingfree.pth"
 - Invoke-WebRequest -Uri
"https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_trainingfree.pth" -OutFile
"checkpoints\MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_trainingfree.pth"
 - Invoke-WebRequest -Uri
"https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_codebook.pkl" -OutFile
"checkpoints\MASt3R_ViTLarge_BaseDecoder_512_catmlpdpt_metric_retrieval_codebook.pkl"

Back to putting this into WSL but also downloading conda onto WSL

```
~/miniconda3/bin/conda init bash
```

Instead of this command:

```
pip install -e thirdparty/in3d
```

Need to do this:

```
git submodule update --init --recursive
```

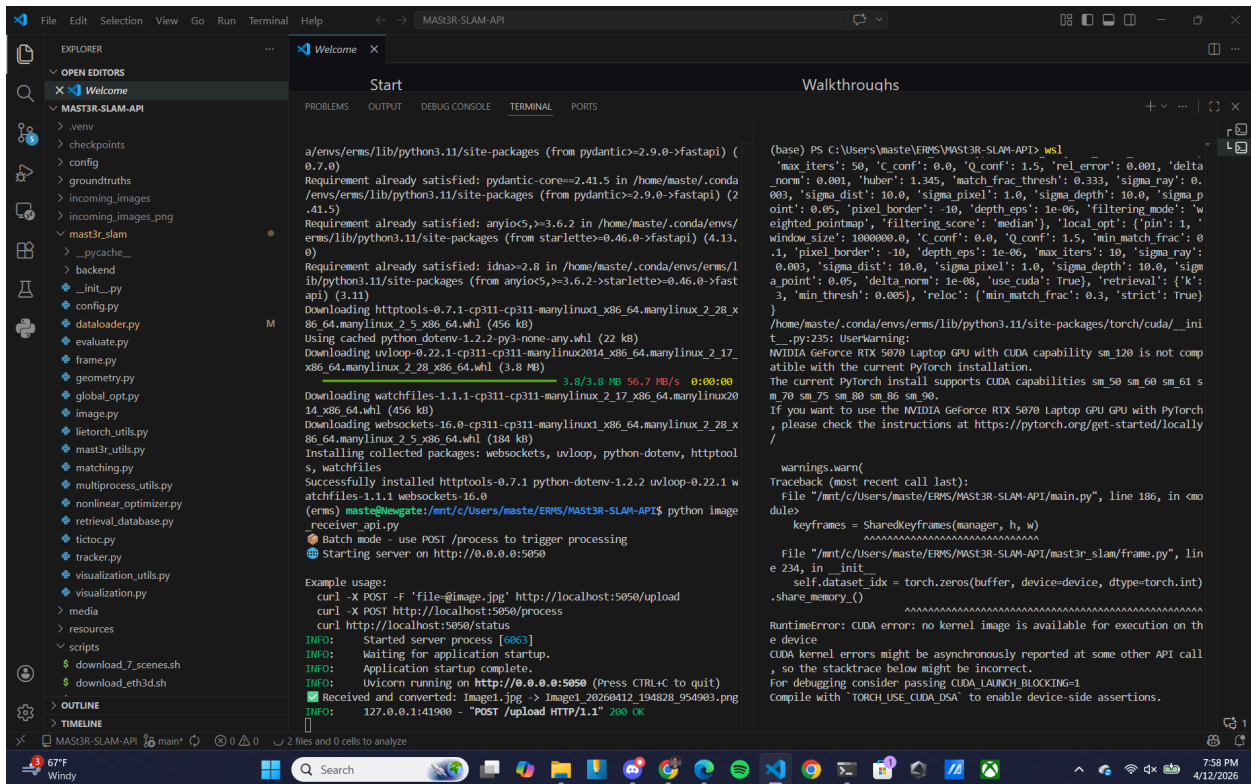
```
pip install -e thirdparty/in3d
```

- Sending images with image_receiver.py
- On a split terminal, send images like this:
curl.exe -X POST -F
"file=@C:\Users\maste\ERMS\MASt3R-SLAM-API\incoming_images\nofloor1.jpg"
<http://localhost:5050/upload>
 - To clear images

```
curl -X POST http://localhost:5050/clear
```

Brandon's Notes

- Issue running python process_images.py



```
File Edit Selection View Go Run Terminal Help
MAS3R-SLAM-API

EXPLORER
OPEN EDITORS
Welcome
MAS3R-SLAM-API
.env
checkpoints
config
groundtruths
incoming_images
incoming_images.png
mast3r_slam
__pycache__
backend
__init__.py
config.py
data_loader.py
evaluate.py
frame.py
geometry.py
global_opt.py
image.py
lietorch_utils.py
mast3r_utils.py
matching.py
multiprocess_utils.py
nonlinear_optimizer.py
retrieval_database.py
ticloc.py
tracker.py
visualization_utils.py
visualization.py
media
resources
download_7_scenes.sh
download_eth3d.sh
OUTLINE
TIMELINE

Start
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

a/envs/erms/lib/python3.11/site-packages (from pydantic>2.9.0->fastapi) (
0.7.0)
Requirement already satisfied: pydantic-core==2.41.5 in /home/maste/.conda
/envs/erms/lib/python3.11/site-packages (from pydantic>2.9.0->fastapi) (2
.41.5)
Requirement already satisfied: anyio<5,>=3.6.2 in /home/maste/.conda/envs/erms/
lib/python3.11/site-packages (from starlette>0.46.0->fastapi) (4.13.
0)
Requirement already satisfied: idna>=2.8 in /home/maste/.conda/envs/erms/l
ib/python3.11/site-packages (from anyio<5,>=3.6.2->starlette>0.46.0->fast
api) (3.11)
Downloading httpx-0.7.1-cp311-cp311-manylinux1_x86_64.manylinux_2_28_x
86_64.manylinux_2_5_x86_64.whl (456 kB)
Using cached python-dotenv-1.2.2-py3-none-any.whl (22 kB)
Downloading uvloop-0.22.1-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_
x86_64.manylinux_2_28_x86_64.whl (3.8 MB)
3.8/3.8 MB 56.7 MB/s 0:00:00
Downloading watchfiles-1.1.1-cp311-cp311-manylinux_2_17_x86_64.manylinux20
14_x86_64.whl (456 kiB)
Downloading websockets-16.0-cp311-cp311-manylinux1_x86_64.manylinux_2_28_x
86_64.manylinux_2_5_x86_64.whl (184 kB)
Installing collected packages: websockets, uvloop, python-dotenv, httpcore
s, watchfiles
Successfully installed httpcore-0.7.1 python-dotenv-1.2.2 uvloop-0.22.1 w
atchfiles-1.1.1 websockets-16.0
(erms) maste@tegate:/mnt/c/Users/maste/ERMS/MAS3R-SLAM-API$ python image
_receiver_api.py
Batch mode - use POST /process to trigger processing
Starting server on http://0.0.0.0:5050

Example usage:
curl -X POST -F 'file=@image.jpg' http://localhost:5050/upload
curl -X POST http://localhost:5050/process
curl http://localhost:5050/status
INFO: Started server process [5063]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:5050 (Press CTRL+C to quit)
INFO: Received and converted: Image1.jpg -> Image1_20260412_194828_954903.png
INFO: 127.0.0.1:41900 - "POST /upload HTTP/1.1" 200 OK

(base) PS C:\Users\maste\ERMS\MAS3R-SLAM-API> ws1
'max_iters': 50, 'C_conf': 0.0, 'Q_conf': 1.5, 'rel_error': 0.001, 'delta
_norm': 0.001, 'huber': 1.345, 'match_frac_thresh': 0.333, 'sigma_ray': 0.
003, 'sigma_dist': 10.0, 'sigma_pixel': 1.0, 'sigma_depth': 10.0, 'sigma_p
oint': 0.05, 'pixel_border': -10, 'depth_eps': 1e-06, 'filtering_mode': 'w
eightd_pointmap', 'filtering_score': 'median', 'local_opt': {'pin': 1, '
window_size': 1000000.0, 'C_conf': 0.0, 'Q_conf': 1.5, 'min_match_frac': 0
.1, 'pixel_border': -10, 'depth_eps': 1e-06, 'max_iters': 10, 'sigma_ray':
0.003, 'sigma_dist': 10.0, 'sigma_pixel': 1.0, 'sigma_depth': 10.0, 'sigm
a_point': 0.05, 'delta_norm': 1e-06, 'use_cuda': True}, 'retrieval': {'k':
3, 'min_thresh': 0.005, 'reloc': {'min_match_frac': 0.3, 'strict': True}
}
/home/maste/.conda/envs/erms/lib/python3.11/site-packages/torch/cuda/_ini
t.py:235: UserWarning:
NVIDIA GeForce RTX 5070 Laptop GPU with CUDA capability sm_120 is not comp
atible with the current PyTorch installation.
The current PyTorch install supports CUDA capabilities sm_50 sm_60 sm_61 s
m_70 sm_75 sm_80 sm_86 sm_90.
If you want to use the NVIDIA GeForce RTX 5070 Laptop GPU with PyTorch
, please check the instructions at https://pytorch.org/get-started/locally
/

warnings.warn(
Traceback (most recent call last):
  File "/mnt/c/Users/maste/ERMS/MAS3R-SLAM-API/main.py", line 186, in <mo
dule>
    keyframes = SharedKeyframes(manager, h, w)
    ~~~~~
  File "/mnt/c/Users/maste/ERMS/MAS3R-SLAM-API/mast3r_slam/frame.py", lin
e 234, in _init
    self.dataset_idx = torch.zeros(buffer, device=device, dtype=torch.int)
    ~~~~~
RuntimeError: CUDA error: no kernel image is available for execution on th
e device
CUDA kernel errors might be asynchronously reported at some other API call
, so the stacktrace below might be incorrect.
For debugging consider passing CUDA_LAUNCH_BLOCKING=1
Compile with 'TORCH_USE_CUDA_DSA' to enable device-side assertions.
```

- The main issue is that my torch is not compatible with my GPU
- Use this command to check

```
python -c "import lietorch; print('lietorch ok')"
```

- Other checks for lietorch
- python - <<'PY'

```
import torch
```

```
import lietorch
```

```
x = lietorch.SE3.Identity(1, device="cuda:0")
```

```
print("lietorch runtime ok", x)
```

```
PY
```

- python - <<'PY'

```
import lietorch
```

```
x = lietorch.SE3.Identity(1, device="cuda:0")
```

```
y = x.inv()
```

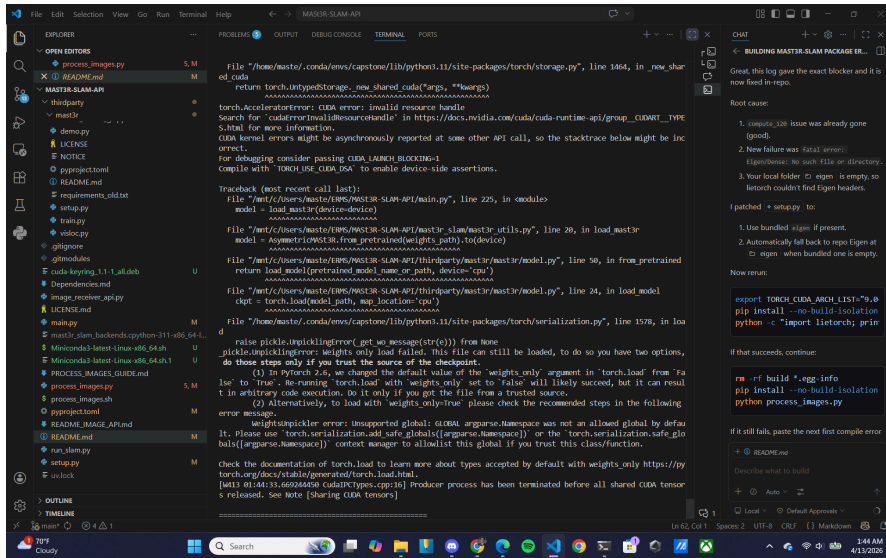
```
print("lietorch cuda runtime ok")
```

```
print(y)
```

```
PY
```

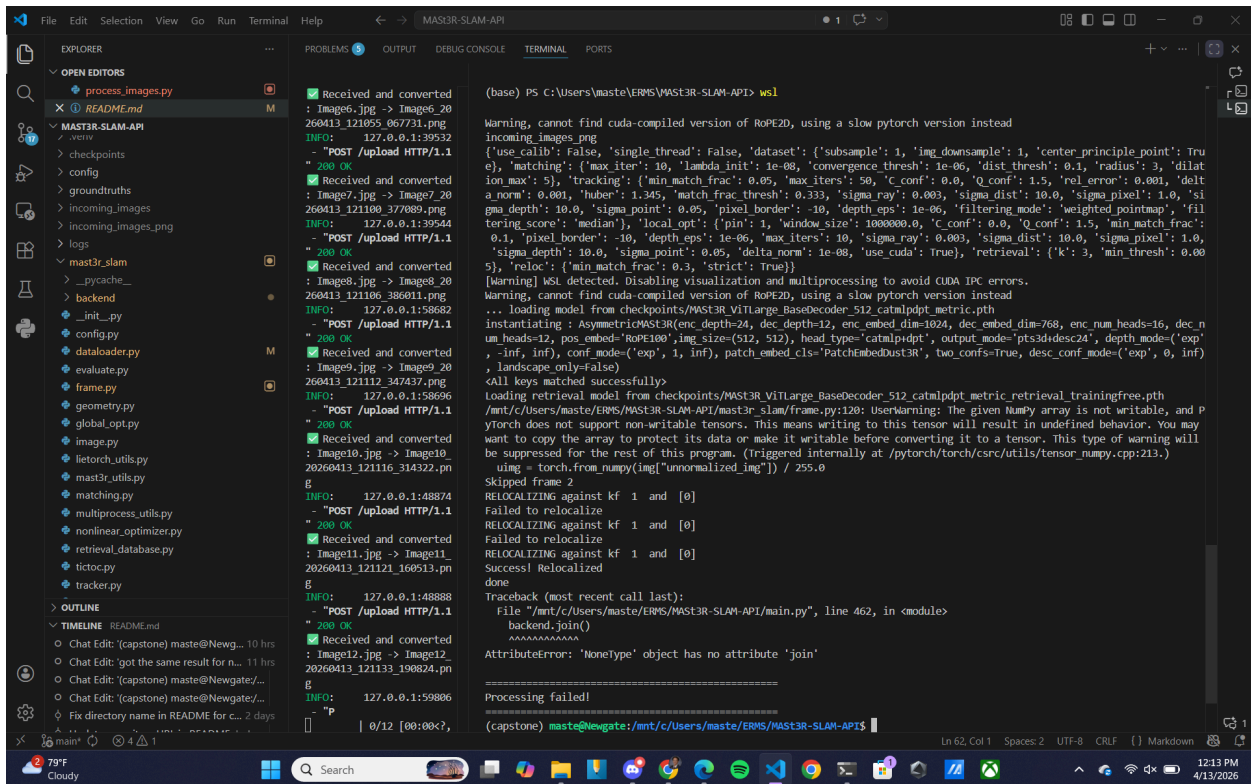
- Still having same problems

Brandon's Notes



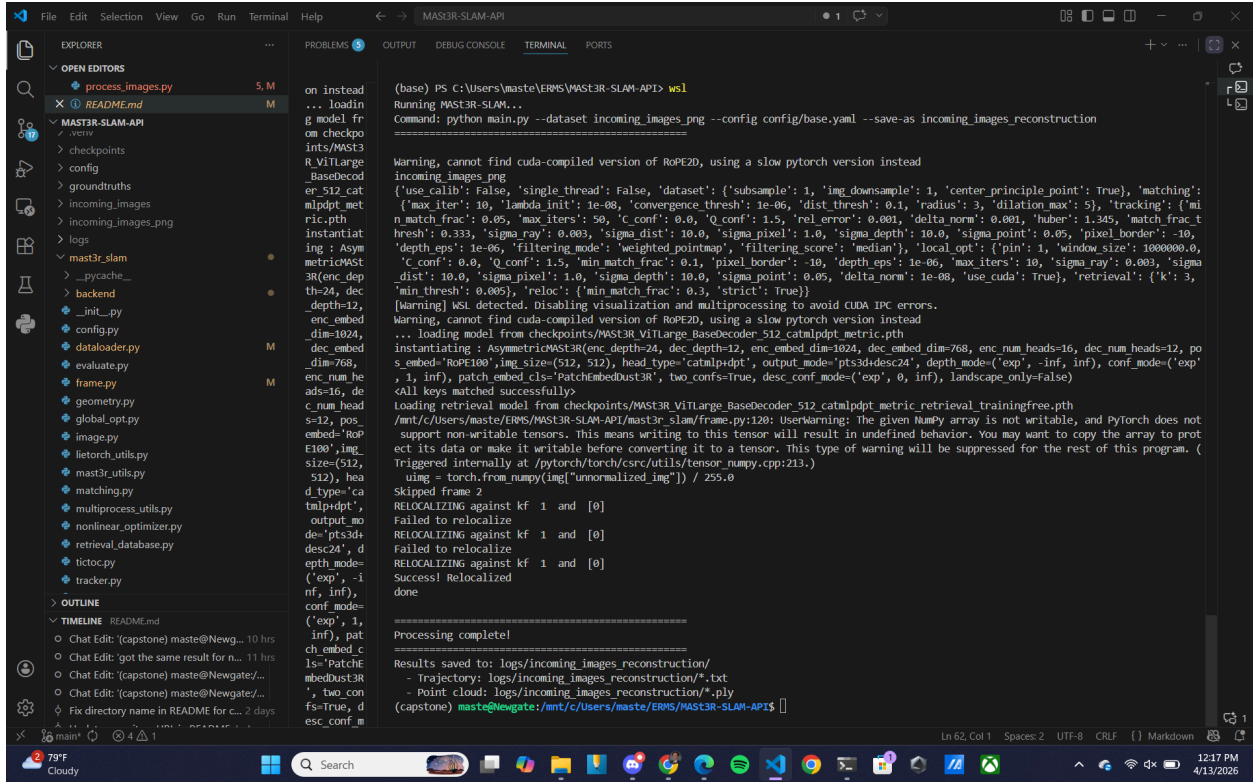
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- Currently having this issue with process_images.py

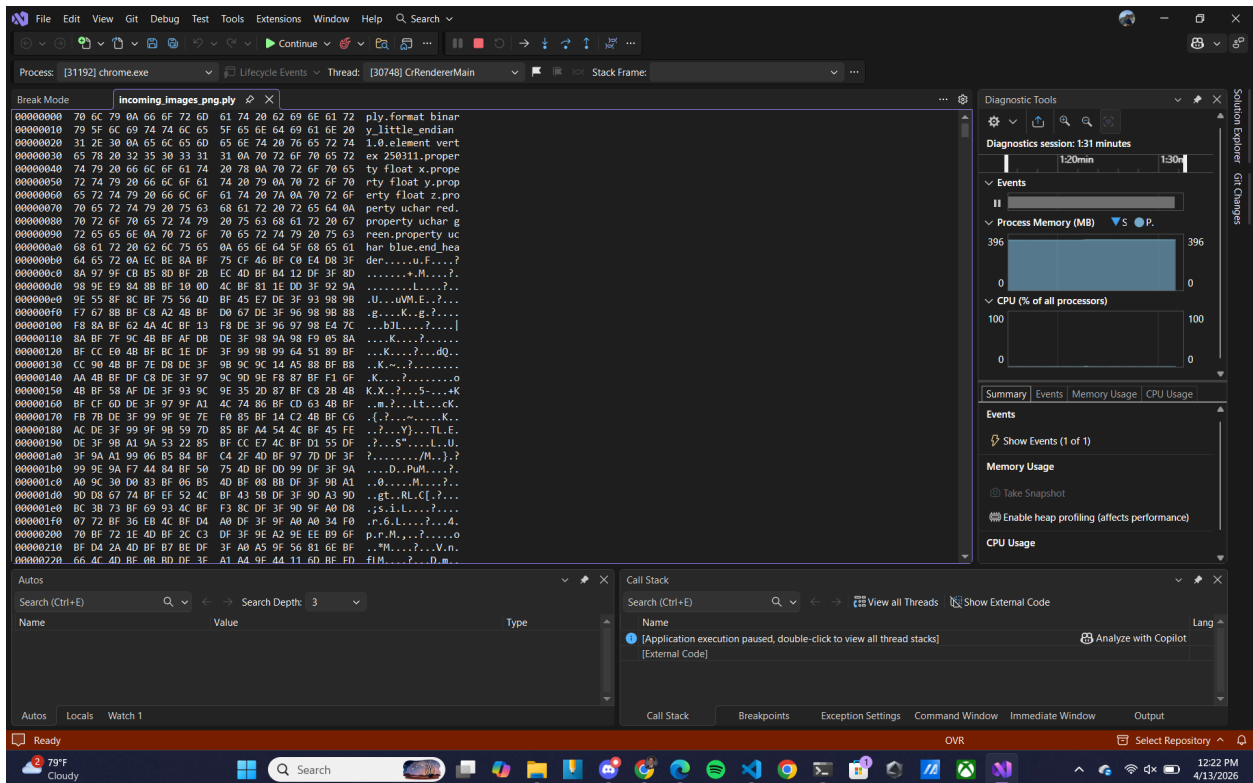


- Fixed some things in main to get this

Brandon's Notes

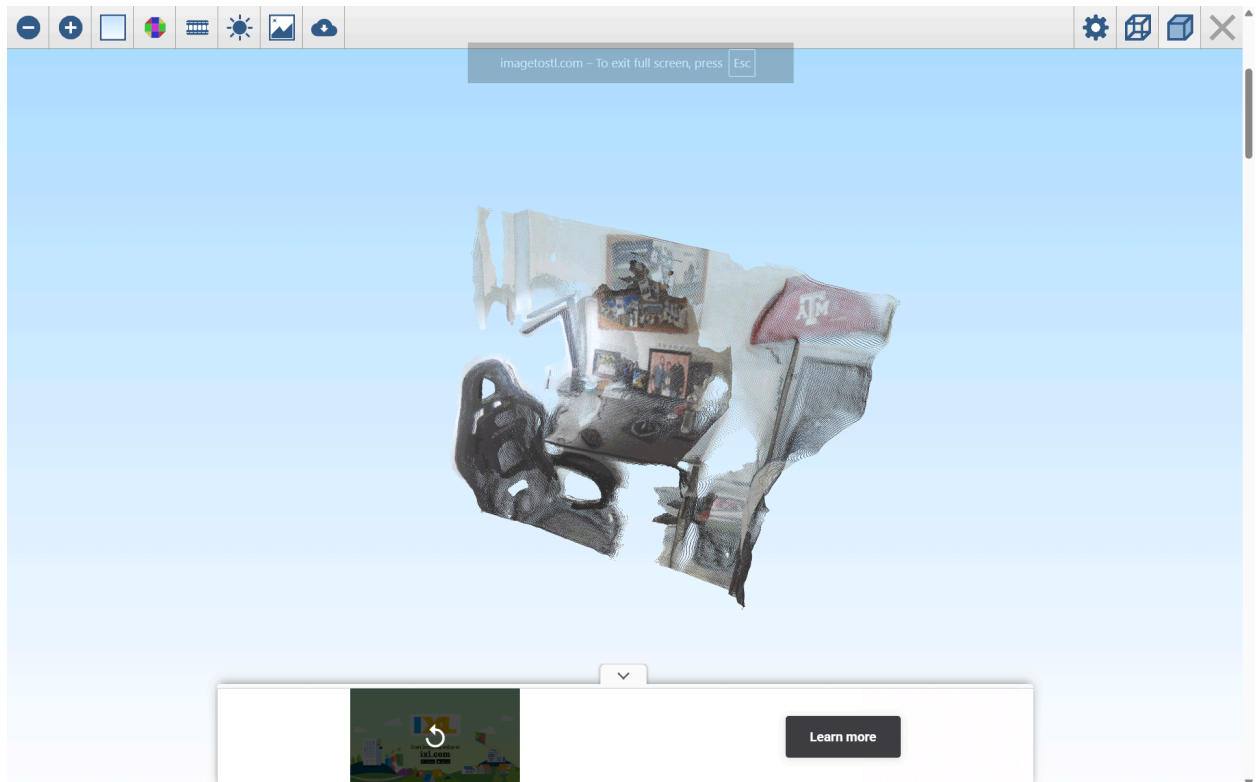


- Now: I must figure out how to view the results on the ply file



Brandon's Notes

- Using this website: <https://imagnetostl.com/view-ply-online#convert>



- For git ignore
- ```
echo "" >> .gitignore
echo "Miniconda3-latest-Linux-x86_64.sh" >> .gitignore
echo "Miniconda3-latest-Linux-x86_64.sh.*" >> .gitignore
echo "cuda-keyring_*.deb" >> .gitignore
git add .gitignore
```

To run the code:

- Wsl
- Conda activate capstone
- Python process\_images.py

4/14 - Meeting with Professor Nima Kalantari

- Akash Garg Room 411 for consultation about ply file viewing
- Recommends 5-10 pictures for each angle of a room → make sure the cameras can take that many photos

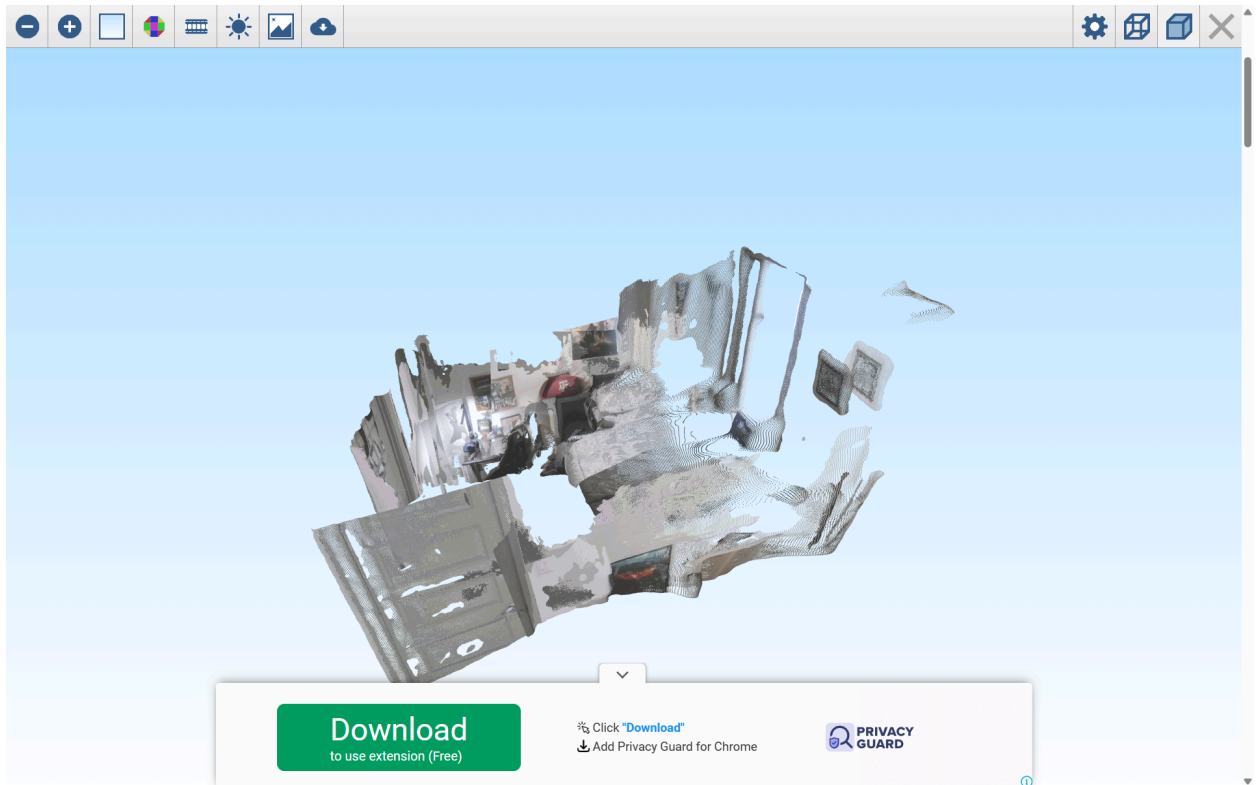
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- Game plan for testing
  - If the ball rolls 5-10 seconds and can only take pictures at 1 fps, we have only about 15 images
  - Did not get any angle with the floor



## Brandon's Notes

- Must capture photos within 1 minute
- Note: If you already put jpg files into incoming\_images, you don't need to upload to the server
- Took ~2 minutes to process process\_images.py with 29 images



- 
- Was able to get more of the surrounding room
- Created a new folder called Old Runs if you want to revisit those ply files

4/20

### Todos

- Takes images at 512x288 resolution to send
- Upload 2304x1296 without downsizing
- Stride downsize 2304x1296 to 512x288, upload 512x288
- Lanczos
- Try to get a model in 30 seconds

Josh's Notes:

It seems as there are only two fovs with keeping the rest of the configuration constant. The minimum resolutions for the narrow and wide angle fovs are 512x288 and 2304x1296 respectively. Then, we only have to test two different fovs.

Considering we have the choice of downsizing the high resolution image, we should consider whether it can really yield a faster upload speed. If downsizing before mast3r slam runs is faster than passing the large image to mast3r slam, the entire process will definitely be faster since the pre-sized image does not have to be downsized by mast3r.

For simplicity, we start with a simple stride approach (take every  $2304/512 = 4.5$ th pixel).

## Brandon's Notes

Image capture is known to take much less time than upload. Then, we should use some parallelism by continuing to take pictures and buffer them as we send them to the server. The ball can continue to upload the images after the ball has stopped rolling, but we want the cameras to always be capturing while still rolling in the center of the room.

Since we are using some kind of parallelism, we should measure the time it takes for the entire use case, not just one image.

| Approach                                             | Latency (processing to upload) |
|------------------------------------------------------|--------------------------------|
| Upload 2304x1296                                     |                                |
| Stride downsize 2304x1296 to 512x288, upload 512x288 |                                |

Since it is faster, we can take the proper approach and try the same downsizing mast3r uses: lanczos resampling.

|                                                       |  |
|-------------------------------------------------------|--|
| Lanczos downsize 2304x1296 to 512x288, upload 512x288 |  |
|-------------------------------------------------------|--|

Pi login

- Hostname: Pizero1
  - Black battery w/ black cable
  - ip: 10.224.32.125
- Hostname: Pizero2
  - pink battery w white cable
  - ip: 10.224.32.42
- Both are username raspberrypi, password raspberry
- Hotspot password: asianyellowant

| Pi Zero 2 W #1                                                                                                                                                                                                                                                                                                                                                                                                 | Pi Zero 2 W #2                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hostname: <a href="#">pizero1</a><br>Localization: <a href="#">&lt;Skip for now&gt;</a><br>Username: <a href="#">raspberrypi</a><br>Password: <a href="#">raspberrypi</a><br>Wi-Fi: <ul style="list-style-type: none"><li>• SSID: <a href="#">Anthill</a></li><li>• Password: <a href="#">asianyellowant</a></li></ul> SSH Authentication: <ul style="list-style-type: none"><li>• <b>Enable SSH</b></li></ul> | Hostname: <a href="#">pizero2</a><br>Localization: <a href="#">&lt;Skip for now&gt;</a><br>Username: <a href="#">raspberrypi</a><br>Password: <a href="#">raspberrypi</a><br>Wi-Fi: <ul style="list-style-type: none"><li>• SSID: <a href="#">Anthill</a></li><li>• Password: <a href="#">asianyellowant</a></li></ul> SSH Authentication: <ul style="list-style-type: none"><li>• <b>Enable SSH</b></li></ul> |



## Brandon's Notes

|                               |                               |
|-------------------------------|-------------------------------|
| • Use password authentication | • Use password authentication |
|-------------------------------|-------------------------------|

|                                                    |                                                   |
|----------------------------------------------------|---------------------------------------------------|
| Hostname: pizero1<br>(black battery + black cable) | Hostname: pizero2<br>(pink battery + white cable) |
| >> <a href="#">ssh raspberrypi@100.88.7.74</a>     | >> <a href="#">ssh raspberrypi@100.94.196.58</a>  |

### Pi Zero 2 W Setup: Tailscale VPN Setup

```
>> curl -fsSL https://tailscale.com/install.sh | sh
>> sudo tailscale up
>> <open authentication link>
>> tailscale ip -4
```

- Problem right now: Can't seem to connect to the Pi via ssh, may need to get an SD card reader to guarantee a connection

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Hostname: pibrandon  
Username: raspberrypi  
Password: raspberry

Wifi Hotspot  
User: Jarvis  
Password: Masterle

IP Address: 172.20.10.3

ssh [raspberrypi@pibrandon.local](#) (takes about 2 minutes to detect from laptop)

Cloned from my branch "brandon" into the Pi  
sudo apt install git -y  
Git clone: <https://github.com/TeamBallers/MASt3R-SLAM-API.git>

Created environment (.env) \  
`source .venv/bin/activate`  
In the pi

Baseline:  
One tab  
`python image_receiver_api.py --continuous --port 5050`

Second tab: (also .venv)  
Cd scripts  
`python3 pi_camera_uploader.py --url http://127.0.0.1:5050/upload --duration 30 --interval 1.0`

## Brandon's Notes

Or

```
python scripts/pi_camera_uploader.py --url http://127.0.0.1:5050/upload
--duration 30 --interval 0.5
```

Check on picture inventory:

```
curl -X POST http://127.0.0.1:5050/clear
curl http://127.0.0.1:5050/status
```

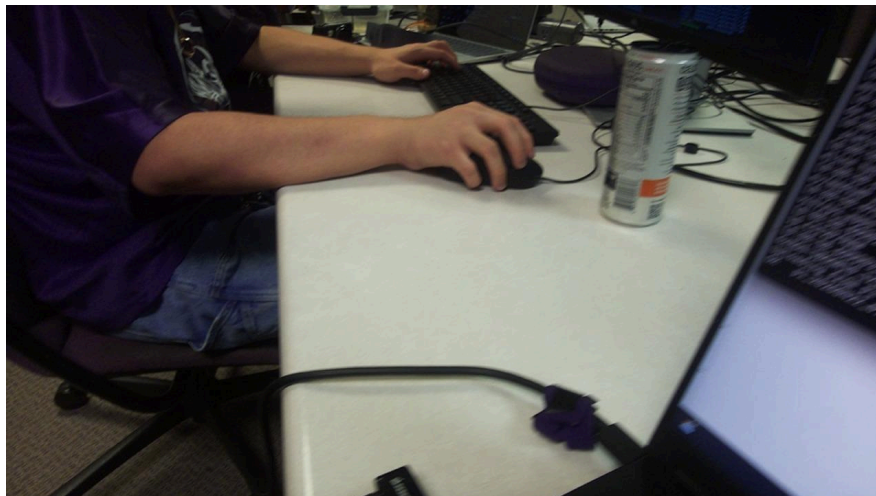
Got 11 images in 30 seconds

```
raspberrypi@pibrandon: ~/M X raspberrypi@pibrandon: ~/M X + v
[0:06:33.150666252] [1175] WARN CameraSensorProperties camera_sensor_properties.cpp:550 Please consider updating the ca
mera_sensor_properties database
[0:06:33.445020519] [1175] INFO IPAPProxy ipa_proxy.cpp:180 Using tuning file /usr/share/libcamera/ipa/rpi/vc4/imx708_wi
de_noir.json
[0:06:33.468390559] [1175] WARN CameraSensor camera_sensor_legacy.cpp:502 'imx708_wide_noir': No sensor delays found in
static properties. Assuming unverified defaults.
[0:06:33.470450083] [1175] INFO Camera camera_manager.cpp:223 Adding camera '/base/soc/i2c0mux/i2c@1/imx708@1a' for pip
eline handler rpi/vc4
[0:06:33.470627053] [1175] INFO RPI vc4.cpp:445 Registered camera /base/soc/i2c0mux/i2c@1/imx708@1a to Unicam device /d
ev/media0 and ISP device /dev/medial
[0:06:33.470797877] [1175] INFO RPI pipeline_base.cpp:1117 Using configuration file '/usr/share/libcamera/pipeline/rpi/
vc4/rpi_apps.yaml'
[0:06:33.504313464] [1170] INFO Camera camera.cpp:1215 configuring streams: (0) 1280x720-BGR888/sRGB (1) 1536x864-SBGR
10_CSI2P/RAW
[0:06:33.505153887] [1175] INFO RPI vc4.cpp:620 Sensor: /base/soc/i2c0mux/i2c@1/imx708@1a - Selected sensor format: 153
6x864-SBGR10_1K10/RAW - Selected unicam format: 1536x864-pBAA/RAW
Uploading from Pi camera to http://127.0.0.1:5050/upload for 30.0s every 0.50s
OK pi_20260422_070533_846880.jpg
OK pi_20260422_070536_697902.jpg
OK pi_20260422_070539_345685.jpg
OK pi_20260422_070542_089125.jpg
OK pi_20260422_070544_701170.jpg
OK pi_20260422_070547_468657.jpg
OK pi_20260422_070550_150363.jpg
OK pi_20260422_070552_853787.jpg
OK pi_20260422_070555_506580.jpg
OK pi_20260422_070558_525568.jpg
OK pi_20260422_070601_220007.jpg
Done. Sent=11, Failed=0
(.venv) raspberrypi@pibrandon: ~/MAST3R-SLAM-API $ |
```

Use this command to put those images into your local folder (not in wsl):

- `scp -r raspberrypi@pibrandon:~/MAST3R-SLAM-API/incoming_images "C:\Users\maste\ERMS\MAST3R-SLAM-API\incoming_images_from_pi"`

It successfully puts photos into a folder

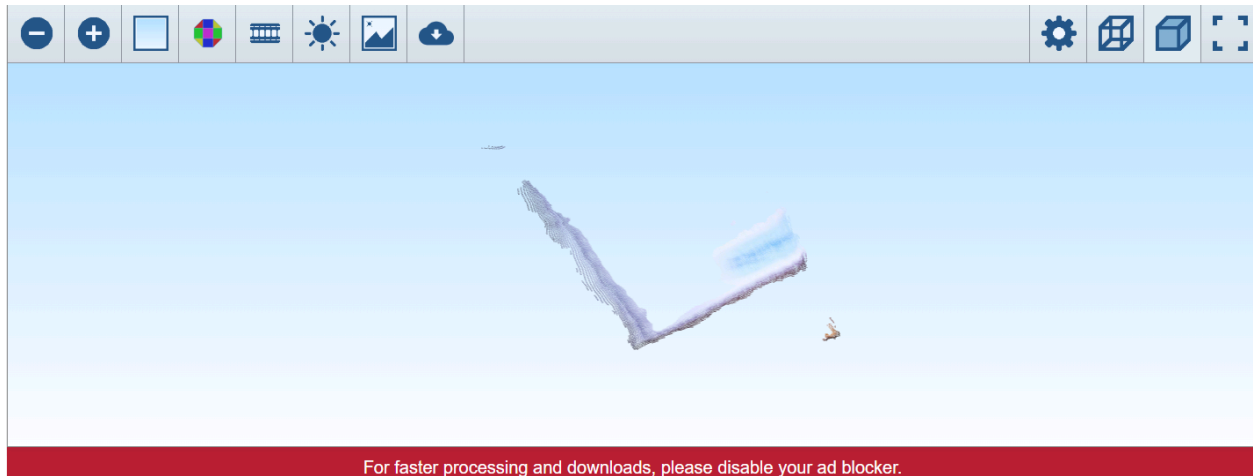


## Brandon's Notes

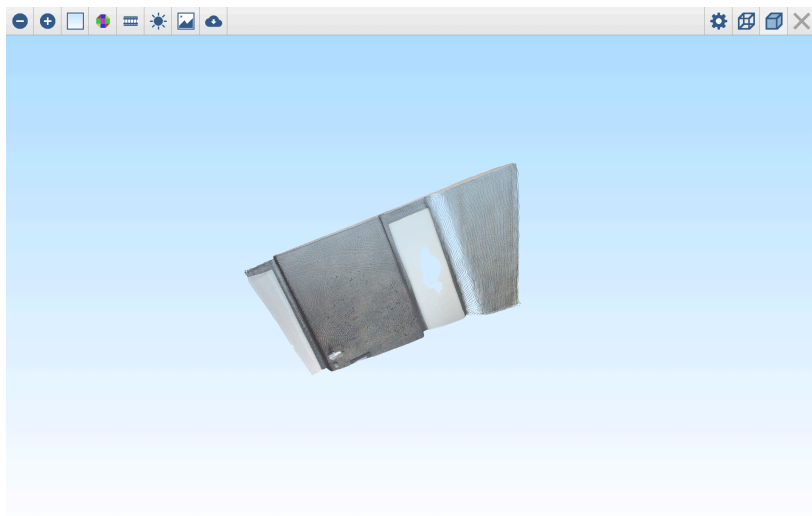
Process in WSL and conda environment on a terminal in VSCode:

```
python process_images.py --input incoming_images_from_pi/incoming_images
--output incoming_images_png --save-as incoming_images_reconstruction
--no-viz
```

360 default



At 512x288 (default to PNG): Kept to one angle at a door with 11 pictures

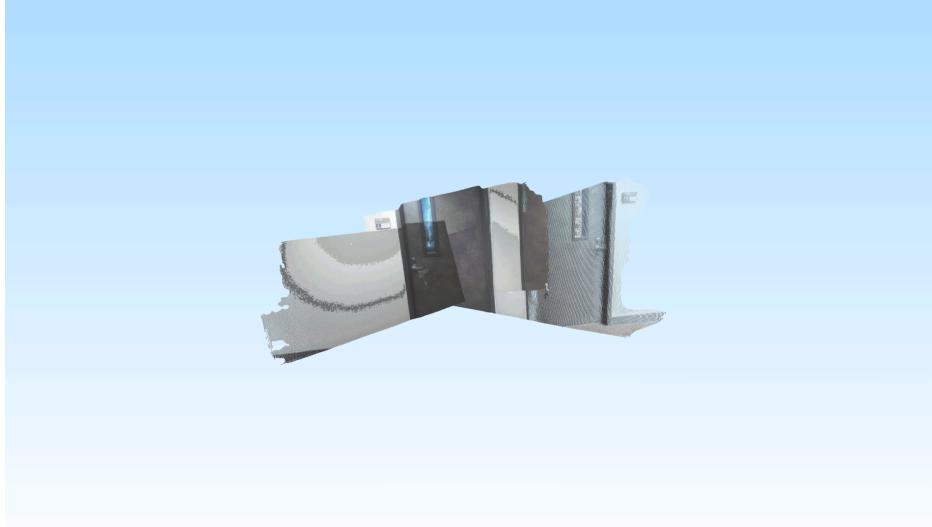


Raw JPG (2304x1296 without downsizing) - 11 images

```
python3 pi_camera_uploader.py raw --duration 30\
```

## Brandon's Notes

```
python process_images.py --input incoming_images_from_pi/incoming_images --fov none
--resize-method none --save-as incoming_images_reconstruction --no-viz
```



Wide (2304x1296 png) - 2 images

```
python3 pi_camera_uploader.py wide --duration 30
```

```
python process_images.py --input incoming_images_from_pi/incoming_images --fov wide
--resize-method mast3r --save-as incoming_images_reconstruction --no-viz
```

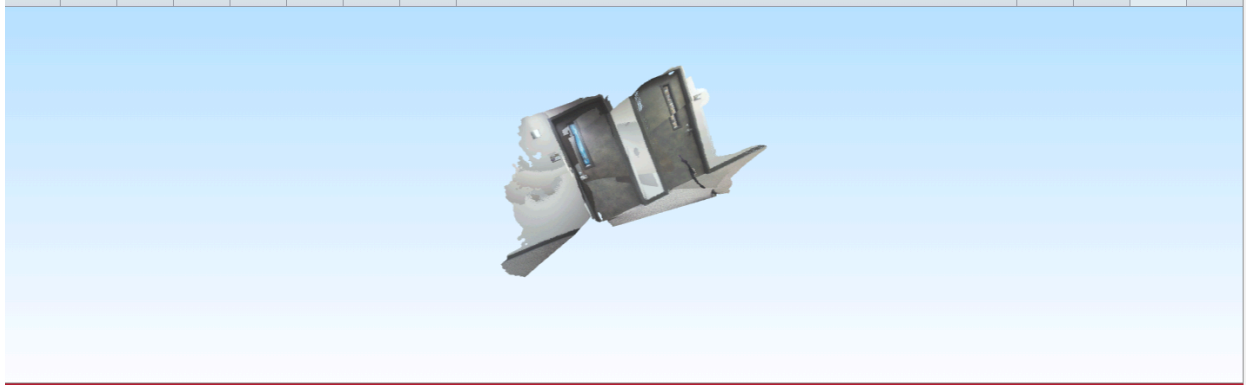


Stride (downsize from 2304x1296 to 512x288) - 22 images

```
python3 pi_camera_uploader.py stride_narrow --duration 30
```

```
python process_images.py --input incoming_images_from_pi/incoming_images --fov narrow
--resize-method stride --save-as incoming_images_reconstruction --no-viz
```

## Brandon's Notes



Lanczos (downsize from 2304x1296 to 512x288) - 15 images

```
python3 pi_camera_uploader.py lanczos_narrow --duration 30
```

```
python process_images.py --input incoming_images_from_pi/incoming_images --fov narrow
--resize-method mast3r --save-as incoming_images_reconstruction --no-viz
```



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Device name: pibrandon

Tailscale IP address:

100.111.255.12

Using the viewer

(used an old test for raw jpg)

## Brandon's Notes

```
python process_images.py --input incoming_images_from_pi/incoming_images --fov
none --resize-method none --save-as incoming_images_reconstruction --no-viz --copy
./ply_viewer/
```

Cd ply\_viewer and python -m http.server 8000

Open link: <http://localhost:8000>



Adjusted commands to ensure the pictures taken can automatically get put into local machine from pi:

(in pi)

```
source .venv/bin/activate
```

(local machine but on wsl)

```
python image_receiver_api.py --pi --continuous --port 5050
```

(pi use tailscale IP)

```
python pi_camera_uploader.py --url http://100.101.179.51:5050/upload --duration 30
--interval 0.5
```

## Brandon's Notes

(local machine but on wsl)

```
python process_images.py --input incoming_images_from_pi/incoming_images --fov none --resize-method none --save-as incoming_images_reconstruction --no-viz --copy ./ply_viewer/
```

Cd ply\_viewer and

```
python -m http.server 8000
```

Open link: <http://localhost:8000>

Check status:

```
curl http://127.0.0.1:5050/status
```

Clear:

```
curl -X POST http://127.0.0.1:5050/clear
```

Check tailscale:

```
curl -v http://100.101.179.51:5050/status
```

If tailscale has connection timeout:

When the Pi times out connecting to the Windows host over Tailscale (`ConnectTimeoutError` on port 5050), the server is running in WSL2 which only listens on `127.0.0.1`. Tailscale traffic arrives on a different interface and never gets forwarded.

Fix (run once in Admin PowerShell on the Windows host):

# 1. Forward all inbound port 5050 traffic to WSL2

```
netsh interface portproxy add v4tov4 listenport=5050 listenaddress=0.0.0.0 connectport=5050 connectaddress=127.0.0.1
```

If doesn't work:

```
Start-Process powershell -Verb RunAs -ArgumentList "-NoProfile -Command `\"netsh interface portproxy add v4tov4 listenport=5050 listenaddress=0.0.0.0 connectport=5050 connectaddress=127.0.0.1; New-NetFirewallRule -DisplayName 'MASt3R-SLAM API 5050' -Direction Inbound -Protocol TCP -LocalPort 5050 -Action Allow -ErrorAction SilentlyContinue; Write-Host 'Done'\""
```



## Brandon's Notes

Run as Administrator PowerShell window (no Start-Process):

```
$wslIp = (wsl hostname -l).Trim().Split(' ')[0]
$wslIp
netsh interface portproxy delete v4tov4 listenaddress=0.0.0.0 listenport=5050
netsh interface portproxy add v4tov4 listenaddress=0.0.0.0 listenport=5050 connectaddress=$wslIp
connectport=5050
Get-NetFirewallRule -DisplayName "MASt3R-SLAM API 5050" -ErrorAction SilentlyContinue |
Remove-NetFirewallRule
New-NetFirewallRule -DisplayName "MASt3R-SLAM API 5050" -Direction Inbound -Protocol TCP
-LocalPort 5050 -Action Allow
netsh interface portproxy show all
```

You should see:

Listen Address 0.0.0.0, Listen Port 5050, Connect Address equal to your WSL IP (not 127.0.0.1).

Then test in order:

1. In WSL with server running: `curl -v --max-time 5 http://127.0.0.1:5050/status`
2. On Windows: `curl.exe -v --max-time 5 http://127.0.0.1:5050/status`
3. On Pi: `curl -v --max-time 5 http://100.101.179.51:5050/status`

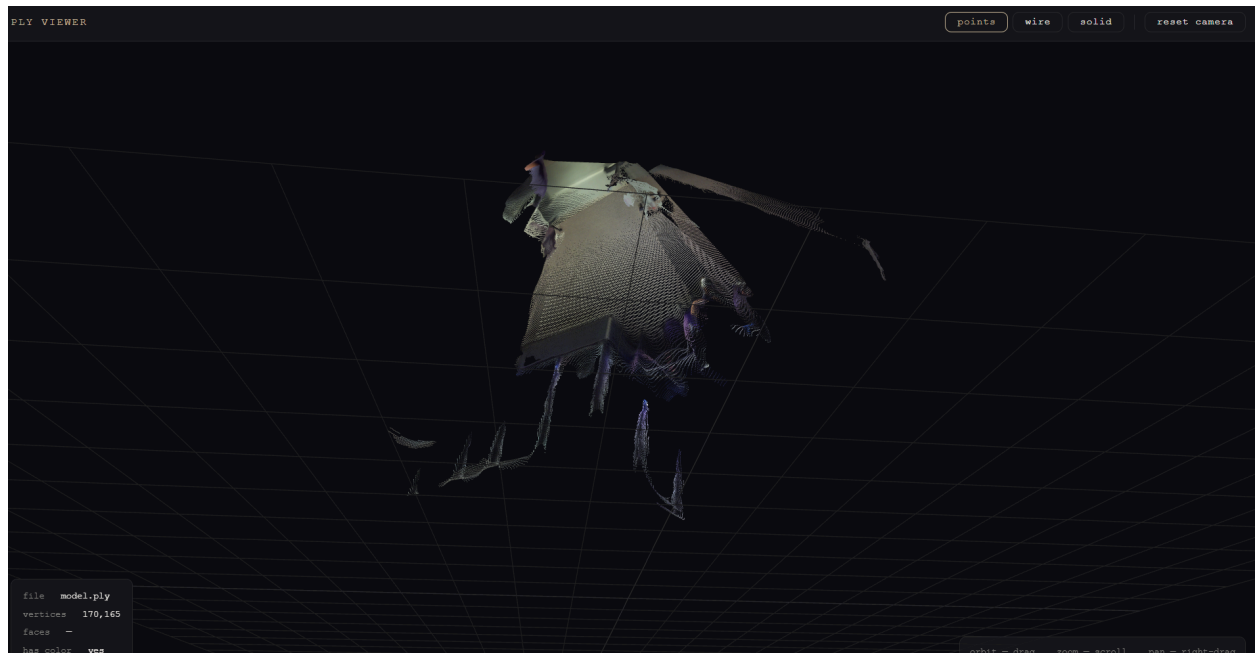
Verify it worked:

```
netsh interface portproxy show all
```

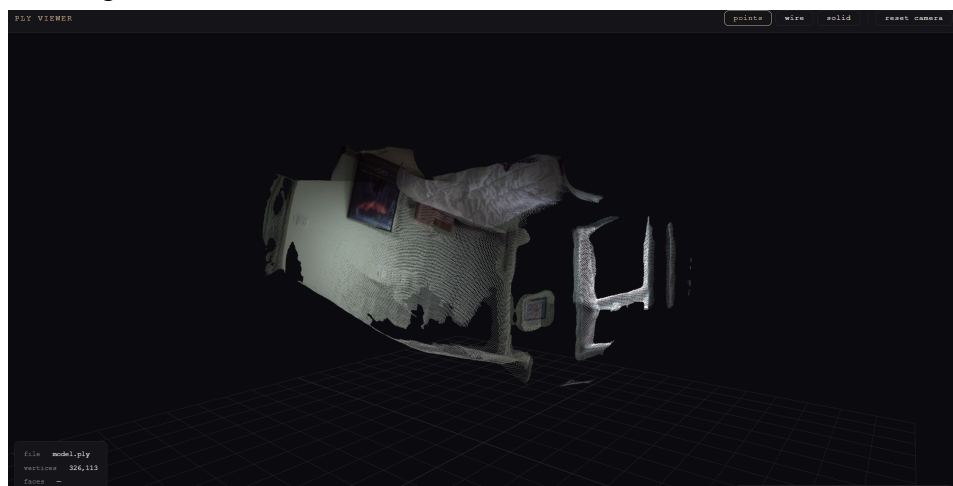
Expected output should show: `0.0.0.0 5050 127.0.0.1 5050`

## Brandon's Notes

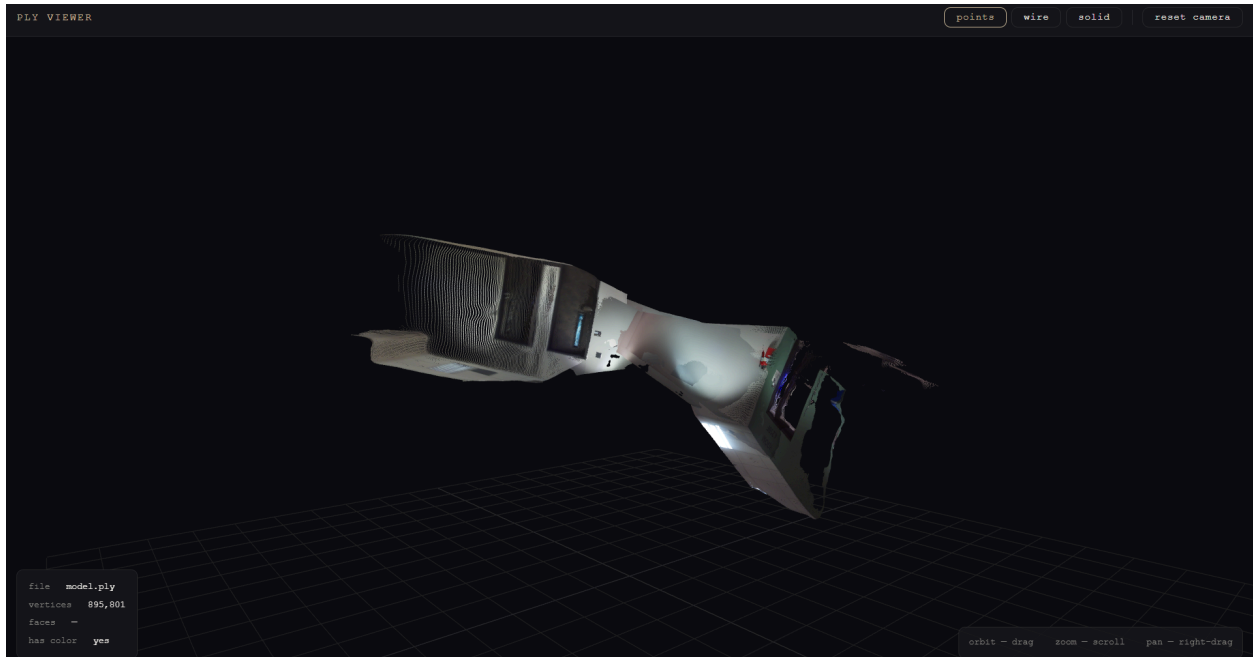
With 7 images using lazcoss



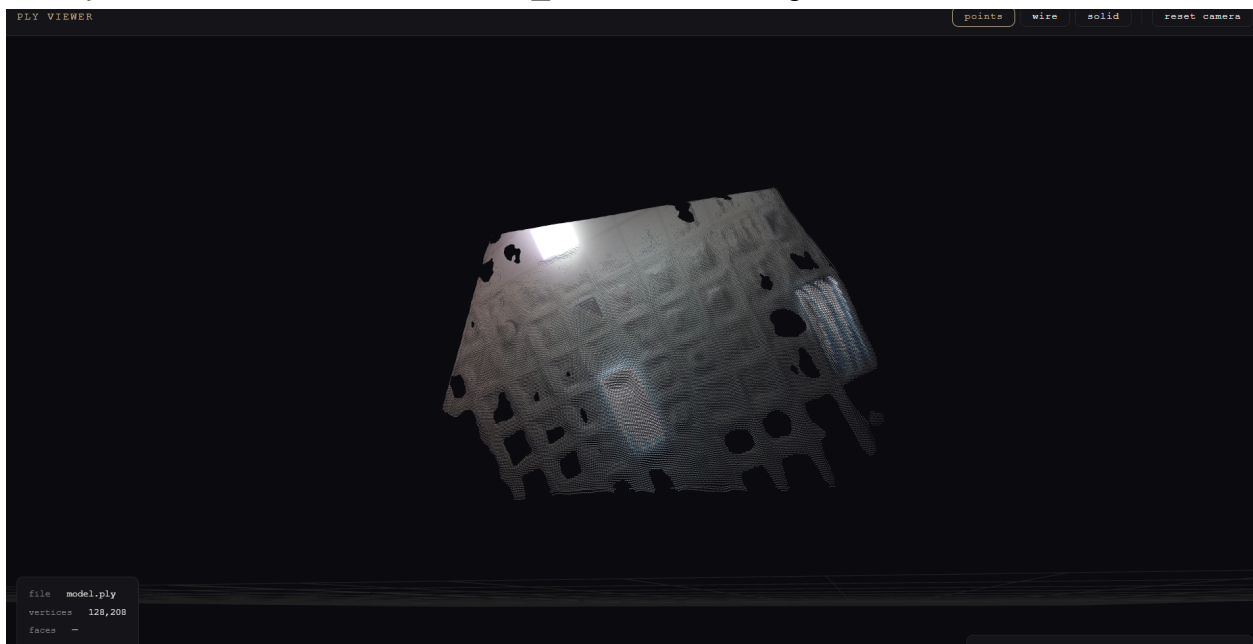
16 images with lazcoss



## Brandon's Notes



Attempted to move around with `stride_narrow` in 15 images



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**Main guide on how to run this**

IP address for pi 4: `ssh raspberrypi@100.86.41.99`

Wsl and conda activate capstone

## Brandon's Notes

Object detection integration with master-slam:

(local machine but on wsl)

```
python image_receiver_api.py --pi --continuous --port 5050
```

```
pip install ultralytics
```

(local)

```
Python yolo_api.py
```

In pi:

```
python camera_client.py --host 100.101.179.51 --port 5050
```

Must manually stop

```
rm runs/api_outputs/annotated/*.jpg
```

In wsl:

```
python process_images.py --input incoming_images_from_pi/incoming_images --fov
none --resize-method none --save-as incoming_images_reconstruction --no-viz --copy
./ply_viewer/
```

If the annotated images have different sizes:

```
python process_images.py --input incoming_images_png --fov narrow --resize-method
lanczos --save-as incoming_images_reconstruction --no-viz --copy ./ply_viewer/
```

Cd ply\_viewer and

```
python -m http.server 8000
```

Open link: <http://localhost:8000>

Check status:

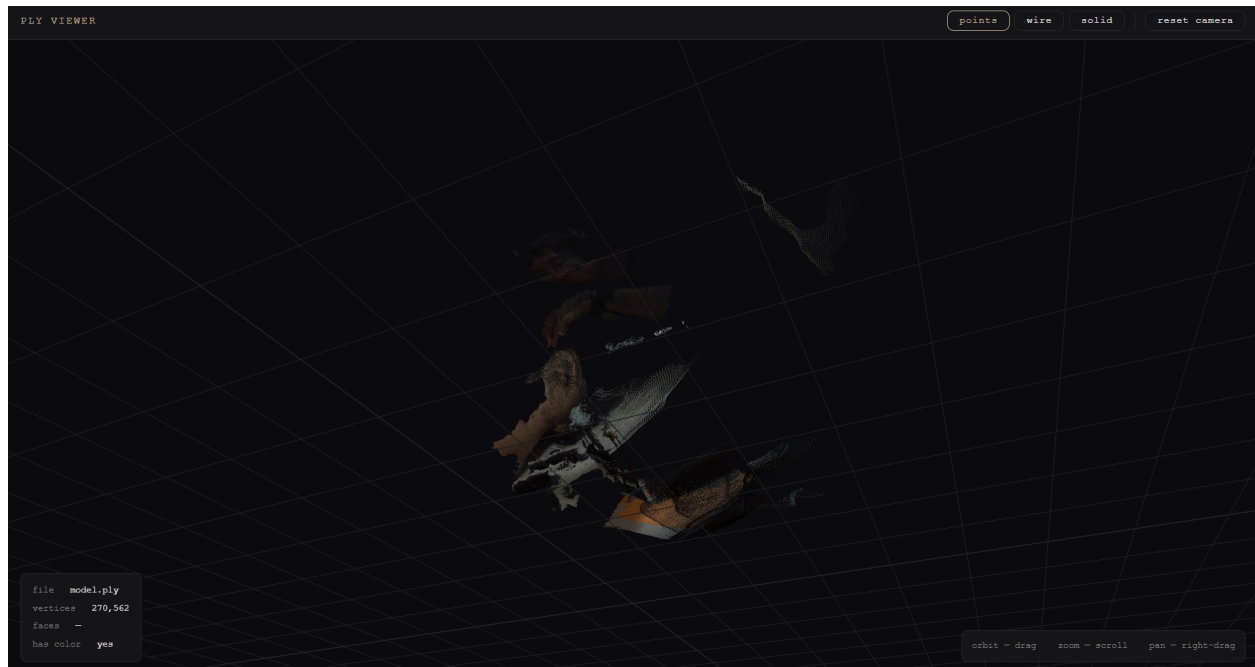
```
curl http://127.0.0.1:5050/status
```

To clear:

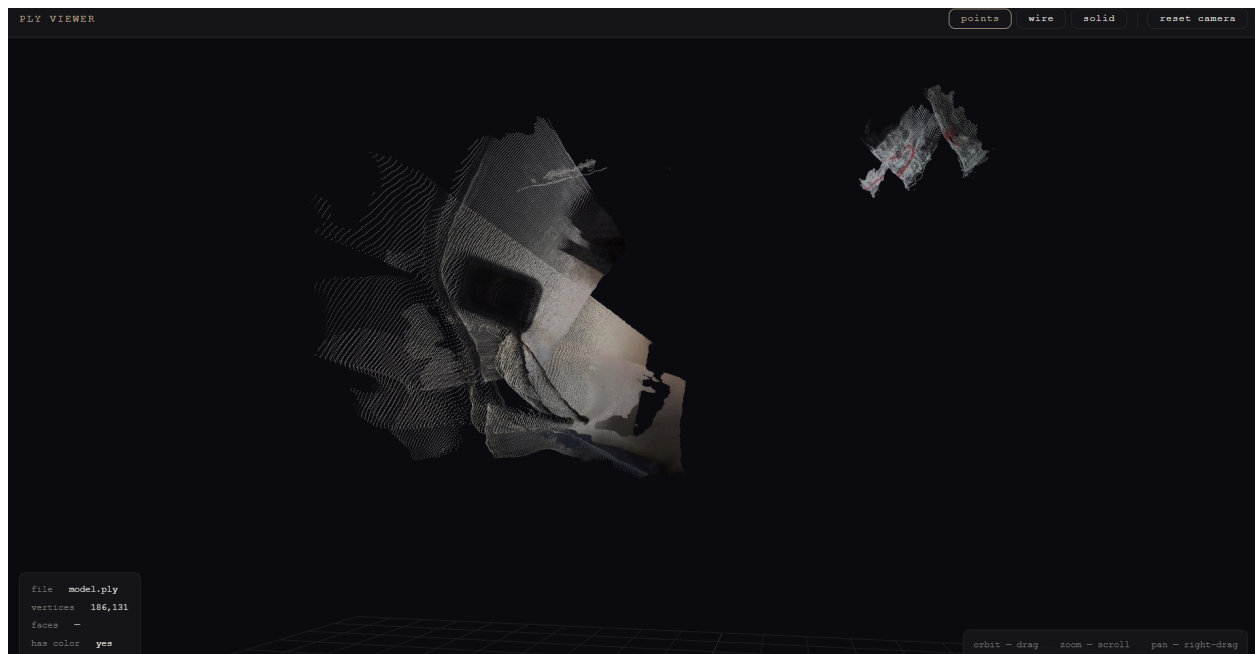
```
curl -X POST http://127.0.0.1:5050/clear
```

Note: be VERY careful with how the camera moves because otherwise the ply just won't create the model. Best to move slowly from side to side

## Brandon's Notes



Test with faster movement and updated camera client



# Set up Guide

<https://github.com/TeamBrowsers/MASt3R-SLAM-API>

# Getting Started

## Installation

```
conda create -n mast3r-slam python=3.11
conda activate mast3r-slam
```

Check the system's CUDA version with nvcc

```
nvcc --version
```

Install pytorch with matching CUDA version following:

```
CUDA 11.8
conda install pytorch==2.5.1 torchvision==0.20.1 torchaudio==2.5.1
pytorch-cuda=11.8 -c pytorch -c nvidia
CUDA 12.1
conda install pytorch==2.5.1 torchvision==0.20.1 torchaudio==2.5.1
pytorch-cuda=12.1 -c pytorch -c nvidia
CUDA 12.4
conda install pytorch==2.5.1 torchvision==0.20.1 torchaudio==2.5.1
pytorch-cuda=12.4 -c pytorch -c nvidia
```

Use mkl < 2025 since that will break installation

```
conda install "mkl<2025.0.0"
```

Clone the repo and install the dependencies.

```
git clone https://github.com/TeamBrowsers/MASt3R-SLAM-API.git --recursive
cd MASt3R-SLAM-API/

if you've clone the repo without --recursive run
git submodule update --init --recursive

pip install -e thirdparty/mast3r # use --no-build-isolation if torch is not
found
pip install -e thirdparty/in3d
pip install --no-build-isolation -e .

Optionally install torchcodec for faster mp4 loading
pip install torchcodec==0.1
```

Setup the checkpoints for MASt3R and retrieval. The license for the checkpoints and more information on the datasets used is written [here](#).

```
mkdir -p checkpoints/
wget
https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_Bas
eDecoder_512_catmlpdpt_metric.pth -P checkpoints/
wget
https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_Bas
eDecoder_512_catmlpdpt_metric_retrieval_trainingfree.pth -P checkpoints/
wget
https://download.europe.naverlabs.com/ComputerVision/MASt3R/MASt3R_ViTLarge_Bas
eDecoder_512_catmlpdpt_metric_retrieval_codebook.pkl -P checkpoints/
```

## WSL Users



We have primarily tested on Ubuntu. If you are using WSL, please checkout to the windows branch and follow the above installation.

```
git checkout windows
```

This disables multiprocessing which causes an issue with shared memory as discussed [here](#).

## Examples

```
bash ./scripts/download_tum.sh
python main.py --dataset datasets/tum/rgbd_dataset_freiburg1_room/ --config
config/calib.yaml
```

## Live Demo

Connect a realsense camera to the PC and run

```
python main.py --dataset realsense --config config/base.yaml
```

## Running on a video

Our system can process either MP4 videos or folders containing RGB images.

```
python main.py --dataset <path/to/video>.mp4 --config config/base.yaml
```

```
python main.py --dataset <path/to/folder> --config config/base.yaml
```

If the calibration parameters are known, you can specify them in `intrinsics.yaml`

```
python main.py --dataset <path/to/video>.mp4 --config config/base.yaml --calib
config/intrinsics.yaml
```

```
python main.py --dataset <path/to/folder> --config config/base.yaml --calib
config/intrinsics.yaml
```

## Downloading Dataset

### TUM-RGBD Dataset

```
bash ./scripts/download_tum.sh
```

### 7-Scenes Dataset

```
bash ./scripts/download_7_scenes.sh
```

### EuRoC Dataset

```
bash ./scripts/download_euroc.sh
```

## ETH3D SLAM Dataset

```
bash ./scripts/download_eth3d.sh
```

## Running Evaluations

All evaluation script will run our system in a single-threaded, headless mode. We can run evaluations with/without calibration:

## TUM-RGBD Dataset

```
bash ./scripts/eval_tum.sh
bash ./scripts/eval_tum.sh --no-calib
```

## 7-Scenes Dataset

```
bash ./scripts/eval_7_scenes.sh
bash ./scripts/eval_7_scenes.sh --no-calib
```

## EuRoC Dataset

```
bash ./scripts/eval_euroc.sh
bash ./scripts/eval_euroc.sh --no-calib
```

# ETH3D SLAM Dataset

```
bash ./scripts/eval_eth3d.sh
```

## Reproducibility

There might be minor differences between the released version and the results in the paper after developing this multi-processing version. We run all our experiments on an RTX 4090, and the performance may differ when running with a different GPU.

## Acknowledgement

We sincerely thank the developers and contributors of the many open-source projects that our code is built upon.

- [MASt3R](#)
- [MASt3R-SfM](#)
- [DROID-SLAM](#)
- [ModernGL](#)

## Citation

If you found this code/work to be useful in your own research, please considering citing the following:

```
@article{mural2024_mast3rslam,
```

```
title={MASt3R-SLAM}: Real-Time Dense {SLAM} with {3D} Reconstruction Priors},
author={Murai, Riku and Dexheimer, Eric and Davison, Andrew J.},
journal={arXiv preprint},
year={2024}.
}
```

## About

MASt3R-SLAM with HTTP API for streaming image processing

## Resources

[Readme](#)

## License

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## Contributors

No contributors

## Languages

Python60.0%

Cuda25.4%

Shell7.6%

C++3.6%

GLSL3.4%

## Suggested workflows

Based on your tech stack



### Python application

Create and test a Python application.



### Python Package using Anaconda

Create and test a Python package on multiple Python versions using Anaconda for package management.



### MSBuild based projects

Build a MSBuild based project.

[More workflows](#)

## Footer

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# Running master slam and yolo guide

IP address for pi 4: ssh raspberrypi@

ssh raspberrypi@100.70.103.119

100.70.103.119

100.78.250.64 rpizero2w1-1

ssh raspberrypi@100.78.250.64

100.118.101.88 rpizero2w2-2

ssh raspberrypi@100.118.101.88

Wsl and conda activate capstone

Object detection integration with master-slam:

(no need to run process\_images)

python image\_receiver\_api.py --pi --continuous --port 5050 --copy ./ply\_viewer/

(for main branch which accounts for resetting image data and if there's enough)

python image\_receiver\_api.py --continuous --copy ./ply\_viewer/model.ply --port 5050

(local)

Cd server

Python yolo\_api.py

(In pi)

python camera\_client.py --host 100.101.179.51 --port 5050

Must manually stop

Cd ply\_viewer and

python -m http.server 8000

Open link: <http://localhost:8000>

Open link: <http://localhost:8080>

Check status:

curl <http://127.0.0.1:5050/status>



To clear:

```
curl -X POST http://127.0.0.1:5050/clear
```

Add manually for testing purposes:

```
curl -X POST -F
```

```
"file=@/mnt/c/Users/maste/ERMS/MASt3R-SLAM-API/incoming_images/nofloor1.jpg"
```

```
http://127.0.0.1:5050/upload
```

Yolo

```
python3 -m venv yolo_env
```

```
source yolo_env/bin/activate
```

```
.\yolo_env\Scripts\activate
```

```
uvicorn yolo_api:app --host 0.0.0.0 --port 8000
```

```
In erms-object-detection/viewer_test
```

```
> cd viewer_tests
```

```
> python -m http.server 8080
```

```
On separate terminal:
```

```
> tailscale serve 8080
```